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Dynamics of a Fluid-Driven Crack in Three Dimensions by the Finite Difference Method

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The finite difference method is applied to the study of the dynamics of a three-dimensional fluid-filled crack excited into resonance by the sudden failure of a small barrier of area ΔS on the crack surface. The impulse response of the crack is examined for various ratios of crack width to crack length and for several values of the crack stiffness $C = (b/\mu)(L/d)$, where b is the bulk modulus of the fluid, μ is the rigidity of the solid, and L and d are the crack length and crack thickness, respectively. The motion of the crack is characterized by distinct time scales representing the duration of brittle failure and the periods of acoustic resonance in the lateral and longitudinal dimensions of the source. The rupture has a duration proportional to the area of crack expansion and is the trigger responsible for the excitation of the crack into resonance; the resonant periods are proportional to the crack stiffness and to the width and length of the crack. The crack wave sustaining the resonance is analogous to the tube wave propagating in a fluid-filled borehole. It is dispersive, showing a phase velocity that decreases with increasing wavelength. Its wave speed is always lower than the acoustic velocity of the fluid and shows a strong dependence on the crack stiffness, decreasing as the stiffness increases. The initial motion of the crack surface is an opening, and the radiated far-field compressional wave starts with a strong but brief compression which has a duration proportional to the crack stiffness and size of the rupture area; the amplitude of this pulse increases with the area of rupture but decreases with increasing stiffness. Flow into the newly created cavity triggers a pressure drop in the fluid, which produces a partial collapse of the wall propagated over the crack surface at the speed of the crack wave. The collapse of the crack surface generates a weak long-period component of dilatation following the compressional first motion in the far-field P wave train; the dilatational component is clearer in the signal from stiffer cracks when seen in the direction of the rupture. The energy loss by radiation is stronger for high frequencies, resulting in a progressive enrichment of the crack response in lower frequencies over the duration of resonance. These source characteristics translate into a far-field signature that is marked by a high-frequency content near its onset and dominated by a longer-period component in its coda. The source duration shows a strong dependence on the fluid viscosity and associated viscous damping at the crack wall.

INTRODUCTION

Hydraulic fracturing offers great potential for the secondary recovery of oil and gas, as well as for the development of geothermal resources. Significant progress has been made recently in the experimental study of hydrofracturing processes at scales ranging from a few centimeters in the laboratory to tens or hundreds of meters in the field, shedding new light on the mechanics of tensile failure and hydrofracture behavior [*Majer et al.*, 1985; *Widrow et al.*, 1985; *Bame and Fehler*, 1986]. The fracturing of rock by forceful fluid injection apparently plays an important role in natural processes also. For example, *Shaw* [1980] argued that fracture mechanisms are responsible for the transport of magma through the lithosphere, while *Julian* [1983] and *Julian and Sipkin* [1985] suggested that hydrofractures due to magmatic intrusion may naturally occur on a large scale at shallow depths in the earth crust. Evidence for tensile crack formation in a natural geothermal environment has also been gathered in the Neovolcanic zone of Iceland by *Foulger and Long* [1984]. Long-period events and harmonic tremor observed on active volcanoes provide yet another body of observations on the behavior of fluid-driven sources. Seismic data [*Riuscetti et al.*, 1977; *Kamo et al.*, 1977; *Latter*, 1979; *Seidl et al.*, 1981; *Schick et al.*, 1982; *Fehler and Chouet*, 1982; *Fehler*, 1983; *Havskov et al.*, 1983; *Malone*, 1983; *McNutt*, 1986; C. S. Weaver and J. E. Shemeta, unpublished manuscript, 1986] *Koyanagi et al.*, 1986 strongly suggest that these two types of events share a common origin and are the result of the excitation of a volcanic conduit triggered by unsteady magma flow or some other form of magmatic instability of a thermal, chemical, or hydraulic nature. According to *Chouet et al.* [1986], harmonic tremor results from the sustained random occurrence of long-period events; in other words, the long-period event can be viewed as the elementary process of tremor and interpreted as the impulse response of the tremor-generating system. Although the triggering mechanisms or forcing functions affecting the flow can be quite varied in the generation of tremor, the dynamics of such source share obvious similarities with those of a hydrofracture.

Along with these data, new studies of the mechanics of propagation of a fluid-filled crack have been made with emphasis on the theoretical behavior of the stress intensity factor under various crack configurations and for different rates of fluid injection, crack geometries, and fluid viscosities [*Abé et al.*, 1976*a*, *b*; *Bui and Parnes*, 1982; *Hayashi and Abé*, 1982, 1983; *Spence and Turcotte*, 1985; *Sammis and Julian*, 1986]. *Aki et al.* [1977] and *Chouet and Julian* [1985] studied the dynamics of propagating two-dimensional fluid-filled cracks using a stress-velocity finite difference method, and *Chouet* [1981, 1982] used the discrete wave number method [*Bouchon*, 1979] to study the elastic radiation in the near field of propagating fluid-filled and dry cracks with various geometries.

The above studies underscore the need for further theoretical analysis of the source dynamics of hydrofractures and their elastic radiation. The object of this paper is to address some of these problems by studying the motion of a three-dimensional fluid-filled crack excited into resonance by the sudden failure

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over its surface of a small barrier of area ΔS. Beyond the direct benefits of a three-dimensional model of the dynamics of a fluid-filled crack in the quantitative study of crack behavior in the engineering applications of hydrofracturing, knowledge of the impulse response of the crack is of critical importance to the understanding of the crack excitation under sustained pressure fluctuations due to forced periodic pumping in man-made fractures [*Widrow et al.*, 1985] or unsteady fluid transport in natural geothermal and volcanic environments [*Chouet et al.*, 1986]. Such analytical studies of the dynamics of a fluid-driven crack are conspicuously lacking in the literature to this date.

The basic assumption of the model is that the failure of the barrier is instantaneous over the area ΔS so that acoustic emission is possible. In their study of the propagation of fluid-driving cracks, *Aki et al.* [1977] showed that in a homogeneous medium, crack growth is limited by the speed with which the driven fluid can follow the crack tip so that the rupture is quasi-static and does not generate seismic waves. Aki et al., however, suggested that sudden failure can occur under excess fluid pressure if either the material strength varies spatially or the dynamic value of the critical stress intensity factor for failure is lower than its static value. *Sammis and Julian* [1986] further investigated this problem and demonstrated that under appropriate geometrical configuration of the cracks, fluid-filled cracks can propagate catastrophically in a homogeneous medium without consideration of dynamic effects and without the need for rapid fluid flow. Their calculations, which were carried out assuming that the fluid is stationary, show that sudden propagation is possible in a homogeneous medium under remote compressive stress when two pressurized cracks approach each other. Sammis and Julian indicated that an asperity or barrier on the crack surface is a particular case of this configuration and that failure of such asperity does occur jerkily even though the fluid may not move. According to their results, the final failure of a strong patch on the crack wall will be instantaneous even if the fluid viscosity were infinite.

Whether the fluid penetrates the crack tip on a time scale that is short, or long, compared to the elastodynamic process is an important question, which bears directly on the propagation characteristics of the fracture. In the first case the crack tip remains open, while in the latter the cohesive force due to ambient pressure around the crack tip tends to pinch it shut. The details of fluid penetration into the crack tip were investigated in a number of studies of stable crack growth created by hydraulic pressurizing [e.g., *Abé et al.*, 1976*a*, *b*; *Bui and Parnes*, 1982; *Spence and Turcotte*, 1985]. For example, using a penny-shaped crack model, *Abé et al.* [1976*a*, *b*] argued that the fracturing fluid cannot penetrate the entire crack when it is expanding, asserting that for a completely filled crack, singularities exist in the equations of motion and continuity at the crack tip. These investigators have assumed an incompletely filled crack to keep the pressure finite as required to obtain values of the stress-intensity factor that are compatible with experimental values for crack extension in rocks. *Bui and Parnes* [1982], on the other hand, demonstrated that the singular pressure at the crack tip arises from the incompatibility of the pure radial flow model used by Abé et al. and the crack geometry near the crack tip. Using exact equations, which include the transverse component of flow velocity parallel to the crack aperture, they showed that the pressure at the crack tip remains finite even when the fluid fills the entire crack space. To keep the dynamic problem tractable, some idealization is required. In this paper, I follow *Aki et al.* [1977] and *Chouet and Julian* [1985] in assuming that the fluid with the excess pressure ΔP fills the newly created crack space. I simulate the failure of the barrier by applying a pressure step ΔP over the area ΔS and account for the initial penetration of the fluid into the void by assuming that the flow into this void can be described by the continuity relation, which states that the volume flow rate into the cavity is equal to the rate of cavity increase. Although I realize that the fluid viscosity is a rate-limiting factor in the rupture process, I do not dwell on the complexities of the crack tip dynamics here. Rather, I assume that the rupture front propagates with infinite velocity and that the pressure step is applied instantaneously over ΔS. Other idealizations used in the model are that (1) the fluid in the crack behaves like a viscous liquid, (2) the liquid is treated as a two-dimensional compressible continuum, (3) the incremental crack extension and resulting flow velocities are so small that the fluid flow in the fracture is everywhere laminar, and (4) the surrounding rock is perfectly elastic, homogeneous, isotropic, and impermeable.

To simulate the dynamics of the propagating fluid-filled crack, I apply a finite difference scheme that includes both particle velocity and stress in the elastic equations of motion and flow velocity and pressure in the equations of fluid dynamics. This scheme is similar to that used by *Madariaga* [1976], *Aki et al.* [1977], *Fehler and Aki* [1978], *Virieux and Madariaga* [1982], and *Chouet and Julian* [1985] in their studies of various crack problems (see *Aki and Richards* [1980, chapter 15] for a review of the method). *Virieux* [1984, 1986] demonstrated the versatility of the velocity-stress finite difference method by applying it to the study of elastic wave propagation in a heterogeneous medium. In his application the density and Lamé coefficients of the medium are specified at each individual node associated with velocity or stress; internal interfaces are not treated explicitly through boundary conditions but are represented naturally by changes in the elastic parameters and density. This discretization allows the inclusion of fluid into the solid through the simple expedient of specifying a Poison's ratio of 0.5 at the appropriate locations in the numerical grid mesh. In other words, the fluid is treated as an elastic solid with zero rigidity. This simple approach avoids the use of the acoustic equation inside the fluid and escapes the complex problem of connecting the liquid and solid along an interface. Although this method is well designed for the treatment of wave propagation problems in mixed solid and liquid media, it is not directly applicable to the problem of a fluid-driven crack since it avoids the formulation of the equations of the fluid dynamics. Indeed, to treat the physics of this problem accurately, it is necessary to include the equations of mass and momentum transfer in the fluid in order to specify the proper conditions for fluid flow into the expanding volume of the crack. For this reason, I avoid the approach of *Virieux* [1984, 1986] in this paper. Instead, I extend the development of *Chouet and Julian* [1985] to the case of a three-dimensional crack and analyze the source behavior in terms of the crack stiffness and viscous damping loss, two dimensionless parameters that quantify the dynamic behavior of the crack and the effect of fluid viscosity on the characteristics of the resonance induced by the failure of the barrier. I follow with an analysis of the far-field compressional wave radiated by the crack and conclude with a discussion of the implications of the model for various observations, includ-

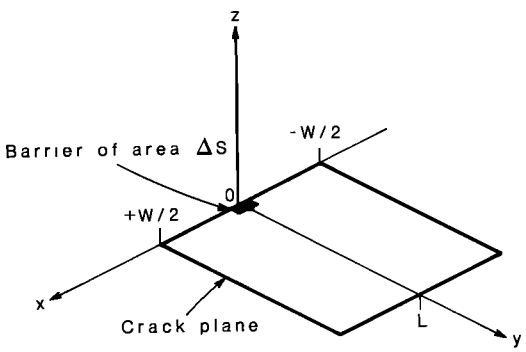

Fig. 1. Sketch of the crack geometry in the Cartesian coordinate system x, y, z considered in the model. The crack is set in the plane $z = 0$ and extends between $x = -W/2$ and $x = +W/2$ in the x direction and from $y = 0$ to $y = L$ in the y direction. The solid rectangle at the origin represents the strong patch of area ΔS on the crack wall, the failure of which triggers the resonance of the crack.

ing the non-double-couple source mechanisms of the 1980 Mammoth Lakes, California, earthquakes [*Wallace et al.*, 1982; *Barker and Langston*, 1983; *Julian*, 1983; *Ekström*, 1983; *Ekström and Dziewonski*, 1983; *Wallace*, 1985; *Julian and Sipkin*, 1985], the source of long-period events in volcanoes [*Fehler and Chouet*, 1982] and man-made hydrofractures [*Bame and Fehler*, 1986], and the source of volcanic tremor [*Chouet et al.*, 1986].

FORMULATION OF THE CRACK PROBLEM

Consider an infinite, linear elastic medium containing a crack whose thickness d is much smaller than the seismic wavelengths of interest. The crack is set in the plane $z = 0$ and extends from $-W/2$ to $+W/2$ along the x axis and from the origin 0 to the distance L in the y direction (Figure 1); it contains a fluid whose pressure exceeds the ambient compressive stress σ_{zz}. At time $t = 0$ a barrier of area ΔS located on the crack perimeter at the origin fails, and the crack suddenly expands by a small incremental volume ΔV, triggering fluid flow into the newly opened cavity. My object is to calculate the motion of the crack surface given appropriate initial conditions over ΔS. Following *Madariaga* [1976], I write the equations of motion of the elastic solid as the following system of nine first-order hyperbolic equations, which propagates explicitly velocity and stress inside the medium

$$\begin{aligned}
\frac{\partial}{\partial t}\sigma_{xx} &= \lambda\left(\frac{\partial \dot{u}}{\partial x} + \frac{\partial \dot{v}}{\partial y} + \frac{\partial \dot{w}}{\partial z}\right) + 2\mu\frac{\partial \dot{u}}{\partial x} \\
\frac{\partial}{\partial t}\sigma_{yy} &= \lambda\left(\frac{\partial \dot{u}}{\partial x} + \frac{\partial \dot{v}}{\partial y} + \frac{\partial \dot{w}}{\partial z}\right) + 2\mu\frac{\partial \dot{v}}{\partial y} \\
\frac{\partial}{\partial t}\sigma_{zz} &= \lambda\left(\frac{\partial \dot{u}}{\partial x} + \frac{\partial \dot{v}}{\partial y} + \frac{\partial \dot{w}}{\partial z}\right) + 2\mu\frac{\partial \dot{w}}{\partial z} \\
\frac{\partial}{\partial t}\sigma_{xy} &= \mu\left(\frac{\partial \dot{u}}{\partial y} + \frac{\partial \dot{v}}{\partial x}\right) \\
\frac{\partial}{\partial t}\sigma_{xz} &= \mu\left(\frac{\partial \dot{u}}{\partial z} + \frac{\partial \dot{w}}{\partial x}\right) \\
\frac{\partial}{\partial t}\sigma_{yz} &= \mu\left(\frac{\partial \dot{v}}{\partial z} + \frac{\partial \dot{w}}{\partial y}\right) \\
\frac{\partial}{\partial t}\dot{u} &= \frac{1}{\rho_1}\left(\frac{\partial}{\partial x}\sigma_{xx} + \frac{\partial}{\partial y}\sigma_{xy} + \frac{\partial}{\partial z}\sigma_{xz}\right) \\
\frac{\partial}{\partial t}\dot{v} &= \frac{1}{\rho_1}\left(\frac{\partial}{\partial x}\sigma_{xy} + \frac{\partial}{\partial y}\sigma_{yy} + \frac{\partial}{\partial z}\sigma_{yz}\right) \\
\frac{\partial}{\partial t}\dot{w} &= \frac{1}{\rho_1}\left(\frac{\partial}{\partial x}\sigma_{xz} + \frac{\partial}{\partial y}\sigma_{yz} + \frac{\partial}{\partial z}\sigma_{zz}\right)
\end{aligned} \tag{1}$$

where λ and μ are the Lamé elastic moduli; ρ_1 is the density of the solid; u, v, and w are the x, y, and z components of displacement, respectively; and σ_{xx}, σ_{yy}, σ_{zz}, σ_{xy}, σ_{xz}, and σ_{yz} are the components of stress. Dots denote time derivatives.

For long wavelengths the motion of the fluid is dominantly in the plane of the crack, and the apparent three-dimensionality of the flow problem can be simplified with little cost in accuracy by treating the fluid dynamics in the fracture as being two-dimensional. The two-dimensional description of the crack fluid dynamics is obtained by integrating the governing equations across the crack aperture. Following *Lamb* [1932], it can be shown [*Wooding*, 1960] that the flow velocity component parallel to the aperture is zero and that the velocity components in the other two directions parallel to the crack plane have a parabolic profile across the aperture at low Reynolds numbers. After integration and averaging over the parabolic velocity profile of the flow, one obtains the equations of conservation of momentum in the x and y directions in the form

$$\begin{aligned}
\frac{1}{\rho_2}\frac{\partial p}{\partial x} + \frac{\partial U}{\partial t} + \frac{12\eta}{\rho_2 d^2}U &= 0 \\
\frac{1}{\rho_2}\frac{\partial p}{\partial y} + \frac{\partial V}{\partial t} + \frac{12\eta}{\rho_2 d^2}V &= 0
\end{aligned} \tag{2}$$

The equation of continuity is given by

$$\frac{1}{b}\frac{\partial p}{\partial t} + \frac{2}{d}\frac{\partial}{\partial t}w_d + \frac{\partial U}{\partial x} + \frac{\partial V}{\partial y} = 0 \tag{3}$$

In these equations ρ_2 is the density, η is the viscosity, and b is the bulk modulus of the fluid; w_d is the normal displacement of the crack wall; p is the fluid pressure; and U and V are the x and y components of the fluid velocity averaged over the flow profile. Convective terms have been neglected in both (2) and (3) under the assumption that these are small compared to the time derivatives of pressure and flow velocity. The viscous shear stresses at the crack wall are related to the average fluid velocities by

$$\begin{aligned}
\tau_x &= \frac{6\eta}{d}U \\
\tau_y &= \frac{6\eta}{d}V
\end{aligned} \tag{4}$$

where τ_x and τ_y represent the x and y components of these stresses, respectively.

The second term in equation (3) expresses the deformation of the crack wall. Note that for a perfectly rigid wall, this term is zero so that when (2) and (3) are combined, the two-dimensional wave equation in terms of pressure is obtained, which for an inviscid fluid reduces to

$$\frac{\partial^2 p}{\partial t^2} = a^2\left(\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2}\right) \tag{5}$$

where $a = (b/\rho_2)^{1/2}$ is the acoustic velocity of the fluid. The elastic yielding of the wall has the effect of lowering the speed of the acoustic disturbance so that the ratio v/a of the crack

wave velocity to the acoustic velocity of the fluid is generally less than 1 [*Chouet and Julian*, 1985]. The velocity of the crack wave $\mathbf{v}$ is typically orders of magnitudes larger than the flow velocities U and V so that the neglect of the convective terms in (2) and (3) is appropriate.

A simple quantitative analysis can show the relative importance of the acceleration and viscous terms in (2). For example, in the water-driven fracture experiments carried out at Fenton Hill, New Mexico, the characteristic, or average, velocity U in the fracture is of the order of 10^{-2} m/s, and the fracture aperture d is of the order of 10^{-3} m [*McFarland and Murphy*, 1976]; the kinematic viscosity of water at 200° (water temperature at the depth of 3500 m) is $\eta/\rho_2 \sim 10^{-7}$ m^2/s [*Schlichting*, 1968], and the frequency of the observed seismic signal is of the order of 100 Hz [*Bame and Fehler*, 1986]. Thus the magnitude of the acceleration term is $\partial U/\partial t \sim 1$ m/s^2, and that of the viscous term is $12\eta U/(\rho_2 d^2) \sim 10^{-2}$ m/s^2. In that case, the flow problem can be solved with good accuracy by assuming that the fluid is inviscid.

At Kilauea volcano, Hawaii, magmatic injection is characterized by a migration velocity of the order of 1 m/s [*Klein et al.*, 1986], the dike aperture is of the order of 1 m [*Swanson et al.*, 1976], the dominant frequency of tremor is of the order of 1 Hz [*Shimozuru et al.* 1966], and the kinematic viscosity of magma in situ is of the order of 10^{-1} m^2/s [*MacDonald*, 1972], so that the acceleration and viscous terms are both of the order of 1 m/s^2, showing that viscous losses cannot be neglected in (2).

At Mount St. Helens, Washington, magmatic injection is characterized by an average velocity of the order of 10^{-2} m/s, and the conduit aperture is of the order of 10^2 m [*Chouet*, 1985], the frequency of long-period events is of the order of 1 Hz [*Fehler and Chouet*, 1982], and the kinematic viscosity of magma in situ is of the order of 10^2 m^2/s [*Chouet*, 1985]. Then the acceleration term has a magnitude of about 10^{-2} m/s^2, while the viscous term is of the order of 10^{-3} m/s^2, indicating that the latter may be neglected in (2). It is of interest that such widely different fluid systems as Mount St. Helens and Fenton Hill should share similarities in their behavior; their common characteristics are demonstrated in the resemblance of the seismic waveforms produced by the two systems (compare, for example, the long-period event in Figure 3 of *Fehler and Chouet* [1982] with those depicted in Figures 2*a* and 2*f* of *Bame and Fehler* [1986]).

The boundary conditions for the problem stated above are found by using the symmetry properties about the crack plane. In the solid the shear stresses σ_{xz}, σ_{yz} and the normal displacement w are odd functions of z and thus vanish over the surface of the crack plane outside the crack. This yields the following boundary conditions for the tensile crack:

$$\begin{aligned} &\sigma_{zz} = -p \qquad \sigma_{xz} = -\tau_x \qquad \sigma_{yz} = -\tau_y \quad \text{on } S_1 \\ &w = 0 \qquad \sigma_{xz} = 0 \qquad \sigma_{yz} = 0 \quad \text{on } S_2 \end{aligned} \tag{6}$$

where S_1 represents the crack surface and S_2 denotes the crack plane outside the crack. The first condition expresses the continuity of normal and shear stresses across the crack face. In addition to the above boundary conditions for the solid, boundary conditions for fluid flow must also be specified at the perimeter of the crack. These are given by

$$\begin{aligned} &U = \dot{u} \qquad \text{along } |x| = W/2 \quad 0 \le y \le L \\ &V = \dot{v} \qquad \text{along } -W/2 \le x \le +W/2 \quad y = 0 \quad y = L \end{aligned} \tag{7}$$

which state the continuity between the peripheral (outflow) component of fluid velocity and the corresponding component of particle velocity in the solid along the crack edges. Note that there are no constraints between the flow velocities U and V and the particle velocities $\dot{u}$ and $\dot{v}$ on the crack surface since these variables are completely defined through (6). In the special case of an inviscid fluid the crack surface is a frictionless boundary, and the fluid is allowed to slip on this surface so that $U \neq \dot{u}$, $V \neq \dot{v}$ on S_1.

I formulate the initial conditions in the same manner as *Aki et al.* [1977] and *Chouet and Julian* [1985] and assume the tensile brittle failure of a small patch of crack of area ΔS. As pointed out by *Aki et al.* [1977], sudden failure can occur when the fluid pressure is high enough and if either the material near the crack tip is heterogeneous or the dynamic value of the critical stress-intensity factor for failure is lower than its static value. In the first case, crack growth can be temporarily interrupted at tough barriers, resulting in unstable crack advance. In the latter, crack growth proceeds uninterrupted once initiated. In the special situation of failure of a barrier on the crack surface that is treated here, the condition for catastrophic rupture is even more favorable. *Sammis and Julian* [1986] demonstrated that instantaneous failure of the patch will occur in this case, independently of the motion of the fluid. In all those situations the sudden expansion of the crack triggers the vibration of the crack wall from which elastic waves are radiated into the solid.

The dynamic motion of the crack surface excited by the crack extension can, in first approximation, be considered as the response of the crack to a sudden step in normal stress σ_{zz} or, equivalently, a pressure step of amplitude ΔP, applied over the area ΔS of the patch. The opening of this cavity triggers fluid flow; following the analysis of *Bui and Parnes* [1982], I assume that the fluid penetrates the entire crack. The initial condition for fluid flow is idealized by the equation of continuity,

$$2\Delta S \dot{w}_d = \dot{Q}_x + \dot{Q}_y \tag{8}$$

where $\dot{Q}_x$ and $\dot{Q}_y$ represent the volume flow rates in the x and y directions. This equation expresses the equality between the rate of fluid flow into the newly formed cavity and the rate of increase of the cavity volume. The step in normal stress over ΔS is given a smal rise time, and (8) is maintained over the duration of this rise time to account in a rough manner for the penetration of the fluid into the cavity. Once this initial condition is satisfied and the volume $2\Delta S w_d$ is occupied by the fluid, the dynamics of this area of the crack are described by (2) and (3) in the same manner as the remainder of the crack. By limiting my study to the case of a small incremental crack extension and assuming instantaneous coupling between the elastic and fluid processes, the physics of the rupture is greatly simplified. In particular, I do not consider the details of the crack geometry near the crack tip, or the effect of fluid viscosity, which acts as a rate-limiting factor in the propagation of the fracture [*Spence and Turcotte*, 1985].

Note that the behavior of the fluid over the expanding region of the crack is treated in a slighly different way here than by *Chouet and Julian* [1985]. In the two-dimensional model of *Chouet and Julian* [1985] the flow into the expanding crack tip is given by (8) together with the condition in the solid $\sigma_{zz} = 2bw_d/d$. These two conditions are maintained as boundary conditions at the crack tip over the entire numerical calculation, while in the present study, (8) is used only

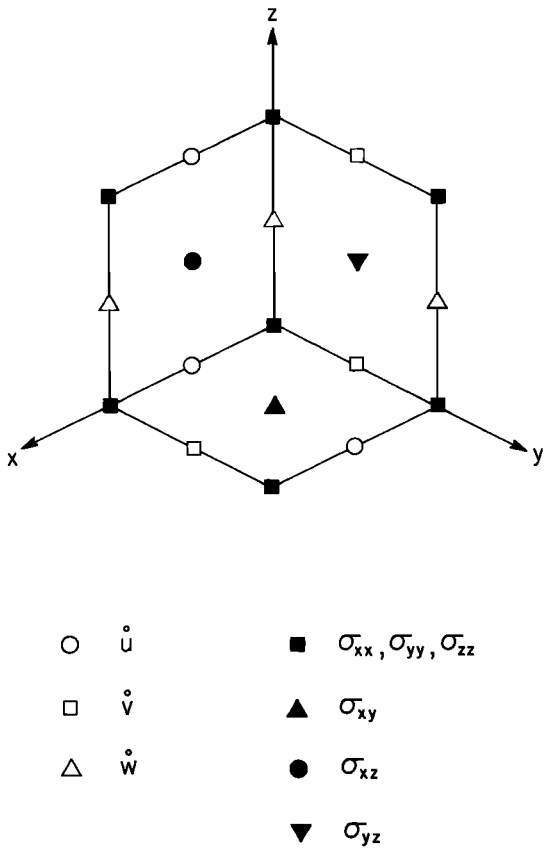

Fig. 2. Finite difference grid used in the computation of the elastodynamic problem.

as an initial condition so that the flow into the newly created cavity is determined in the same manner as the flow over the rest of the source by solving the equations of motion and continuity once the initial filling of this cavity has been accounted for. This approach simulates the fluid dynamics in the expanded area of the crack more closely.

NUMERICAL SIMULATION

The equations of motion (1), (2), and (3) constitute a set of first-order hyperbolic equations for stresses, pressure, and velocities which can be solved numerically by centered finite differences. The method has been described in details by *Madariaga* [1976], *Aki et al.* [1977], *Fehler and Aki* [1978], *Virieux and Madariaga* [1982], *Virieux* [1984, 1986], and *Chouet and Julian* [1985]. My approach follows that of *Aki et al.* [1977] and *Chouet and Julian* [1985]. I simulate the brittle failure of the crack by applying a pressure step ΔP at the appropriate grid location. The resulting opening induces fluid flow which is initially described by (8); the pressure step ΔP is given a short rise time, and (8) is maintained for the duration of this rise. Using the finite difference method, I compute particle velocities and the time derivatives of stresses, from which displacements and stresses are obtained by integration. The initial static field prior to the expansion of the crack is not considered. Rather, the solution represents the opening of the crack relative to its original thickness prior to its extension. As in the work by *Virieux and Madariaga* [1982], I use a numerical grid that requires the fewest number of grid points per finite difference cell. The grid for the solid is displayed in Figure 2. Defining $\Delta x = \Delta y = \Delta z = \Delta$ as the grid spacing, the velocity $\dot{u}$ is obtained at the position $[(i + \frac{1}{2})\Delta, j\Delta, k\Delta]$, the velocity $\dot{v}$ at $[i\Delta, (j + \frac{1}{2})\Delta, k\Delta]$, the velocity $\dot{w}$ at $[i\Delta, j\Delta, (k + \frac{1}{2})\Delta]$, the stresses σ_{xx}, σ_{yy}, σ_{zz} at $[i\Delta, j\Delta, k\Delta]$, the stress σ_{xy} at $[(i + \frac{1}{2})\Delta, (j + \frac{1}{2})\Delta, k\Delta]$, the stress σ_{xz} at $[(i + \frac{1}{2})\Delta, j\Delta, (k + \frac{1}{2})\Delta]$, and the stress σ_{yz} at $[i\Delta, (j + \frac{1}{2})\Delta, (k + \frac{1}{2})\Delta]$. The velocities are estimated at $t = n\Delta t$ and the stresses at $t = (n + \frac{1}{2})\Delta t$. Thus the grid is completely staggered in space and time and has a side-centered cubic symmetry. Because of the mixed boundary conditions (6), I cannot specify $\dot{w}$, σ_{xz}, σ_{yz}, and σ_{zz} on the same numerical plane. As the boundary conditions for $\dot{w}$, σ_{xz}, and σ_{yz} are specified at $z = \Delta/2$, I use their antisymmetry property about the crack plane. This yields $\dot{w}[i\Delta, j\Delta, \Delta/2] = -\dot{w}[i\Delta, j\Delta, -\Delta/2]$, $\sigma_{xz}[(i + \frac{1}{2})\Delta, j\Delta, \Delta/2] = -\sigma_{xz}[(i + \frac{1}{2})\Delta, j\Delta, -\Delta/2]$, and $\sigma_{yz}[i\Delta, (j + \frac{1}{2})\Delta, \Delta/2] = -\sigma_{yz}[i\Delta, (j + \frac{1}{2})\Delta, -\Delta/2]$. On the crack surface itself the constraints given in (6) are met in an approximate way. The stress σ_{zz} is specified at $z = 0$ but depends on the velocity $\dot{w}$ (equation (3)), which is only known on the plane located at $z = \Delta/2$. Similarly, the stress σ_{xz} is defined at $z = \Delta/2$ but depends on the stress τ_x known at $z = 0$, and the stress σ_{yz} is related to τ_y in the same fashion (note that when the fluid is inviscid, τ_x and τ_y both vanish on the crack surface; rather than forcing σ_{xz} and σ_{yz} to vanish at $z = \Delta/2$ in that case, these stresses are set to zero at $z = 0$ through the use of the antisymmetry relations stated above). The offsets between σ_{zz} and $\dot{w}$, σ_{xz} and τ_x, and σ_{yz} and τ_y represent the basic limitation in the resolution of the numerical solution of the fluid-driven crack. However, as none of these variables is singular on the plane $z = 0$, this should not present much of a problem.

The grid for the flow problem is set up in a similar manner as for the solid. I let $\Delta x_a = \Delta y_a = \Delta_a$, where the subscript a refers to the acoustic grid for the fluid. Then, the fluid pressure p is obtained at the position $[l\Delta_a, m\Delta_a]$, the flow velocity U at $[(l + \frac{1}{2})\Delta_a, m\Delta_a]$, and the flow velocity V at $[l\Delta_a, (m + \frac{1}{2})\Delta_a]$. The calculation proceeds with the fluid pressure at $t = (n + \frac{1}{2})\Delta t$ estimated from the fluid pressure at $t = (n - \frac{1}{2})\Delta t$ and the flow velocities and crack opening velocity at $t = n\Delta t$. The calculation again combines the pressure and velocity equations into a single explicit two-step finite difference scheme.

The condition for numerical stability in the solid is given by [*Alford et al.*, 1974; *Virieux and Madariaga*, 1982]

$$H \leq \frac{1}{\sqrt{3}\alpha} \tag{9}$$

where α is the P wave velocity in the solid and $H = \Delta t/\Delta$ is the ratio of the time increment to the grid interval for the elastodynamic problem. There is a similar condition for the fluid, given by

$$H_a \leq \frac{1}{\sqrt{2}a} \tag{10}$$

where a is the acoustic wave velocity in the fluid and $H_a = \Delta t/\Delta_a$ is the ratio of the time increment to the grid interval for the flow problem. Choosing $H = 1/(4\alpha)$ and $H_a = 1/(2a)$ provides the following relations between the two problems

$$H_a = \frac{2\alpha}{a} H \qquad \Delta_a = \frac{a}{2\alpha} \Delta \tag{11}$$

To satisfy this requirement under the condition $a \leq \alpha$, I interpolate between the values of $\dot{w}[i\Delta, j\Delta, \Delta/2]$ using the method

of *Wiggins* [1976]. This approach is similar to that used by *Chouet and Julian* [1985] in the treatment of the two-dimensional fluid-filled crack, although these authors solved the fluid dynamics by the method of characteristics rather than using the hyperbolic partial differential equations of motion treated here. The method of characteristics is a very useful technique for the treatment of two-dimensional problems, yielding ordinary differential equations that are easy to handle numerically; the application of this technique to three-dimensional problems is not straightforward, however, and I did not pursue this approach in this study.

The factors H and H_a are closely related to the numerical dispersion created by the discretization of the wave equation. Precision itself depends on the number of grids chosen to discretize the crack; this is especially true of the tensile crack problem treated here, for the crack stiffness always includes the elasticity of layers of rock, one half grid interval thick, on either face of the crack. This cannot be prevented because the mixed boundary conditions specifying the crack problem relate $w[i\Delta, j\Delta, \Delta/2]$ located a half grid space away from the crack surface to $\sigma_{zz}[i\Delta, j\Delta, 0]$ in the crack plane (see equations (3) and (6) and discussion above). The discretization of the rupture is a source of numerical oscillations. The problem of precision is most severe at the onset of the computation due to the reduced number of grids involved near the initial disturbance front. To palliate these problems, a viscous dissipation term is added to the equations for particle velocities in the solid and flow velocities in the fluid. This term has the form $c\nabla^2 v_i$, where v_i represents the particle velocity in the solid or flow velocity in the fluid and c is a nondimensional attenuation coefficient chosen in such manner as to dissipate the short-wavelength components of the solution as the calculation progresses [*Richtmyer and Morton*, 1967; *Virieux and Madariaga*, 1982]. Various tests made with different values of c indicate that when a large amount of pseudoviscosity is used, the crack response shows a slow rise time, and the profile of the wall opening is smooth behind it; whereas when a small amount of pseudoviscosity is used, the initial rise time is sharper, but there are oscillations of the wall behind it. In choosing a value of c, one must compromise between the desideratea of a sharp onset of crack opening at the barrier and a smooth profile of the wall response behind it. In all my calculations this coefficient is assigned a value of 0.025 in the solid and 0.075 in the fluid. These values represent the optimal amount of pseudoviscosity satisfying the above compromise in the numerical solution of the fluid-driven crack problem.

Because the grid spacing in the fluid is smaller than the grid spacing in the solid when $a < 2\alpha$ (see equation (11)), the perimeter of the crack in the hydraulic problem does not coincide exactly with that defined by the elastic mesh, and there is a difficulty in satisfying the boundary conditions (7) precisely. Rather than forcing these conditions by extrapolation, I assume $U = 0$ and $V = 0$ along the perimeter of the hydraulic mesh. This approximation will need more testing in future work.

The numerical solution is accurate to within a 10% error in the phase velocity for wavelength $\lambda > 5L/n$, or wave number $k = 2\pi/\lambda < 2n\pi/(5L)$, where n is the number of grid points over the length L of the crack. Various tests of the validity and convergence of the leap-frog finite difference scheme can be found in the works by *Madariaga* [1976], *Fehler and Aki* [1978], *Virieux and Madariaga* [1982], and *Chouet and Julian* [1985].

Several approaches are possible to avoid parasitic reflections generated at the boundaries of the numerical mesh. *Chouet and Julian* [1985] avoid the effect of these spurious reflections by specifying a number of grids large enough so that no waves generated at the boundaries arrive back at the crack during the time interval of interest. This simple approach is adequate when one is interested only in the early part of the crack response and the source time window considered is short. Typically, the number of grids between the barrier position and the closest boundary of the numerical mesh for the solid is given by $m \geq \alpha T/(2\Delta)$, where T represents the length of the time window over which the crack response is sought. This approach was selected in the present study when analyzing the details of the initial motion of the crack surface (see Figures 4–11). This simplification, however, rapidly leads to prohibitive computational expenditure when the source response is sought over time windows that encompass the entire history of crack excitation triggered by the failure. In that case, it is necessary to formulate transparent boundary conditions along the outer surfaces, edges, and apexes of the computational mesh to speed up the numerical code. Various approaches are possile to implement these conditions [e.g., *Lysmer and Kuhlemeyer*, 1969; *Smith*, 1974; *Engquist and Majda*, 1977; *Clayton and Engquist*, 1977; *Reynolds*, 1978]. In my study of the resonance characteristics of the source (Figures 12–16), the nonreflecting boundary conditions are implemented by using simple impedance relations based on the assumption of plane P and S waves impinging on the boundary at normal incidence. For example, at the top boundary of the mesh the three variables of interest are $\dot{w}$, σ_{xz}, and σ_{yz} (see Figure 2), which can be expressed in terms of known variables on the adjacent lower plane by the relations $\dot{w} = -\sigma_{zz}/(\rho_1\alpha)$, $\sigma_{xz} = -\rho_1\beta\dot{u}$, and $\sigma_{yz} = -\rho_1\beta\dot{v}$ (here, β represents the shear wave velocity in the solid). Similar relations apply to the front face of the mesh in the positive x direction and to its sides in the positive and negative y directions. These conditions were found to be efficient in removing spurious reflections because in the present application the source effectively localizes the intensity of the wave field to the vicinity of the crack so that parasitic reflections are of second order (Figure 14 demonstrates the source strength in relation to the near field). Tests performed using those impedance relations showed results that matched perfectly with those obtained using the first method discussed above. It must be noted, however, that while simple impedance relations are adequate for the study of the source dynamics in a homogeneous medium, these are not sufficient if one is interested in the propagation of wave fields in a complicated heterogeneous structure because in that case the parasitic reflections are of the same order as the reflections at real interfaces within the medium (see, for example, *Reynolds* [1978] and *Virieux* [1984, 1986]).

The numerical results are expressed as dimensionless functions of the crack and medium parameters through the following scaling:

$$\begin{aligned}
&\text{Length} && r' = r/L \\
&\text{Time} && t' = \alpha t/L \\
&\text{Stress} && \sigma_{ij}' = \sigma_{ij}/\sigma_0 \\
&\text{Displacement} && u_i' = u_i\mu/(L\sigma_0) \\
&\text{Velocity} && \dot{u}_i' = \dot{u}_i\mu/(\alpha\sigma_0)
\end{aligned} \tag{12}$$

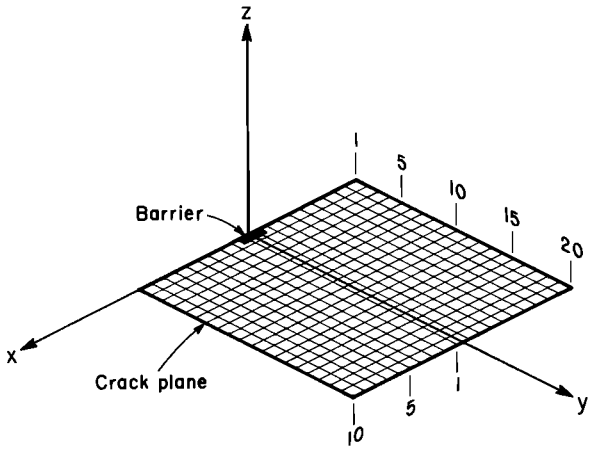

Fig. 3. Discretization of the crack in the Cartesian coordinate system x, y, z. The crack is set in the plane $z = 0$, is symmetrical about the plane $x = 0$, and extends over 20 grid intervals in the x and y directions ($W/L = 1$). The excitation of the crack wall is triggered by the sudden failure of a barrier two grid steps wide along the x axis and one grid step wide in the y direction, shown by the dark patch located at the origin. The numbers refer to individual profiles in the x and y directions discussed in the text.

where nondimensional variables are indicated by a prime. The physical variables are the crack length L, the effective stress σ_0, and the compressional velocity α, and rigidity μ of the solid. For consistency with previous two-dimensional numerical studies of fluid-filled cracks [*Aki et al.*, 1977; *Fehler and Aki*, 1978; *Chouet and Julian*, 1985] I have chosen the particle displacements and velocities to scale with μ rather than $\lambda = 2\mu$. Needless to say, the choice of scaling parameters is arbitrary, and an equivalent solution is obtained whether the shear modulus or another modulus of interest is used.

CRACK EXCITATION TRIGGERED BY THE FAILURE OF THE BARRIER

To alleviate the problem of large computer storage capacity inherent to three-dimensional numerical solutions, I consider here a crack problem that has symmetry about the y axis of the crack. The problem I study, depicted in Figure 3, is that of a barrier with dimensions of two grids along the x axis and one grid in the y direction, located midway along the crack edge $|x| \leq W/2$, $y = 0$, on a crack of length L and width W. The crack shown in Figure 3 is 20 grids long and 20 grids wide ($W/L = 1$). Because the initial conditions are applied over two grids symmetrically about the y axis, the problem has symmetry about the plane $x = 0$, and the solution reduces to that of calculating the wave field in the quarter space $x \geq 0$, $-\infty \leq y \leq +\infty$, $z \geq 0$. The shear stresses σ_{xy} and σ_{xz} and the velocities $\dot{u}$ and U are odd functions of x and thus vanish in the plane $x = 0$, providing the boundary conditions in that plane.

The results of the numerical simulation are presented in Figures 4–12 and the parameters of the various models are summarized in Table 1. The rise time of the pressure step ΔP applied initially over ΔS is fixed to two time steps, and a Poisson ratio of $1/4(\lambda = \mu)$ is used for the solid throughout the calculations. As in the works by *Aki et al.* [1977] and *Chouet and Julian* [1985], the solutions depend critically on a dimensionless parameter called the crack stiffness and defined by

$$C = \frac{b}{\mu}\frac{L}{d} \tag{13}$$

which characterizes the dynamic behavior of the crack. Another dimensionless parameter F, defined by

$$F = \frac{12\eta L}{\rho_2 d^2 \alpha} \tag{14}$$

characterizes the viscous damping loss due to viscous fluid flow within the crack.

The dynamic response of a dry crack ($C = 0$) to the instantaneous failure of a barrier such as illustrated in Figure 3 is shown in Figures 4 and 5. Figure 4 depicts snapshots of the normal component of the particle velocity on the surface of the crack taken at constant intervals of 20 time steps and starting 20 time steps after the failure (one time step represents the interval $\Delta t' = \alpha \Delta t/L = 1/80$ in that model). The barrier is located at the center of the crack edge facing the observer. The amplitude in each individual snapshot has been normalized to a maximum value of unity for a better viewing of the motion of the crack wall, so those plots are not representative of the energy loss by elastic radiation into the solid. Figure 5 shows the normal displacement of the crack surface sampled at various locations over the half of the crack corresponding to $x > 0$ (see Figure 3). Individual profiles parallel to the y axis and numbered 1–10 with increasing x are marked at the top of each stack of seismic traces. There are 20 seismic traces per profile, numbered 1–20 in the positive y direction, as indicated at the left of the figure. Displacement and time are given in dimensionless units of the medium and crack parameters.

Figure 4 illustrates well the propagation of the elastic disturbance over the crack surface and its reflections off the edges of the crack, first from the left and right boundaries $|x| = W/2$ and then from the edge opposite the observer at $y = L$. Later-

TABLE 1. Crack Parameters

Figures	Number of Points	Stiffness C	Viscosity F	Ratio b/μ	Ratio L/d	Ratio ρ_2/ρ_1
4, 5	20 × 20	0	...	...	...	...
6, 7, 20	20 × 20	15	0	1/2	30	2/3
8, 21	20 × 20	50	0	1/2	100	2/3
9, 10, 12b, 13, 15, 16, 22	20 × 40	100	0	1/2	200	2/3
11	20 × 100	250	0	1/2	500	2/3
12a, 24, 25, 26	20 × 40	100	0.1*	1/2	200	2/3
14	40†	5	0	1/10	50	2/15
23	20 × 40	5, 50, 100	0	1/2	10, 100, 200	2/3

*$F = 0.1$ implies a kinematic viscosity $\eta/\rho_2 = 10^{-1}$ m²/s ($\eta = 2500$ P; $\rho_2 = 2.5$ g/cm³) if one assumes $L = 100$ m, $d = 0.5$ m, and $\alpha = 3000$ m/s.

†Two-dimensional crack model.

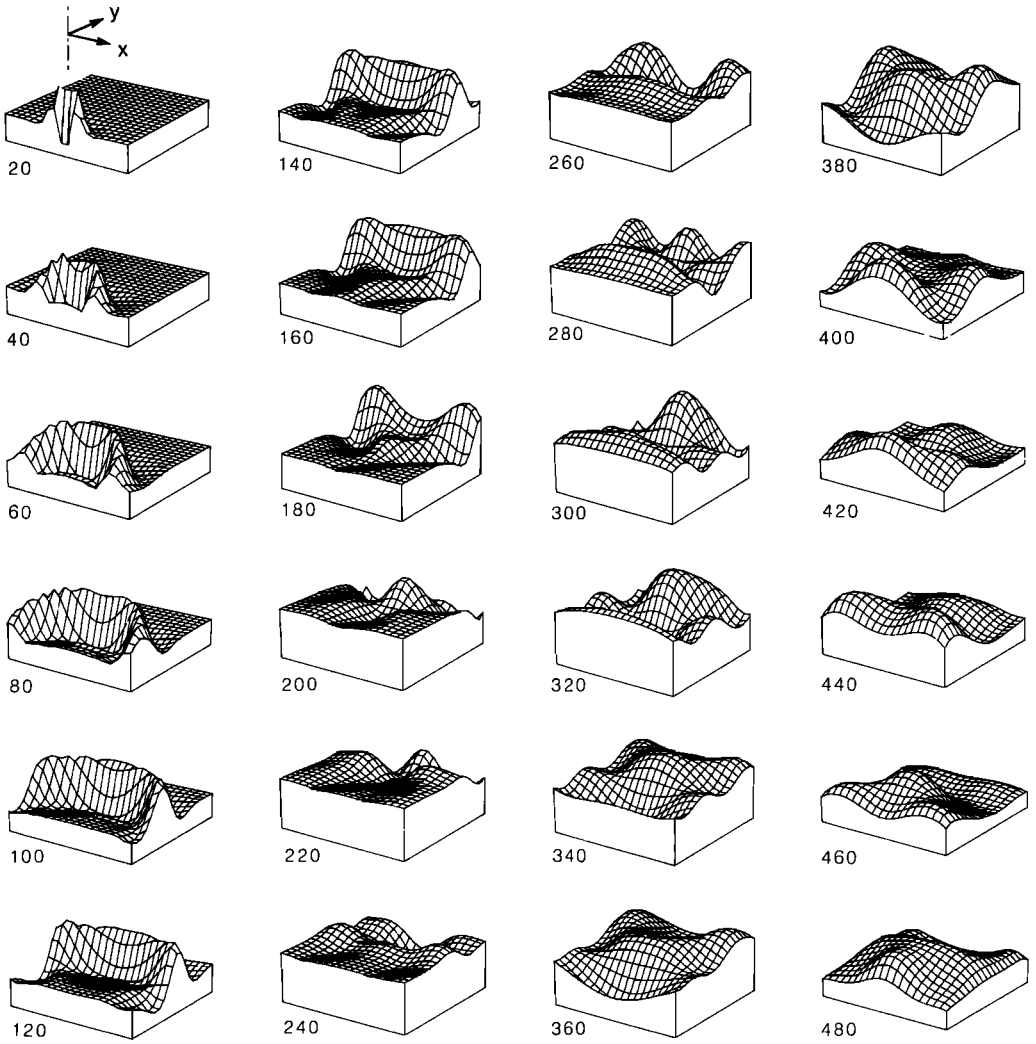

Fig. 4. Snapshots of the normal component of the particle velocity on the surface of the crack for a dry crack excited by the failure of a barrier located at the center of the front edge of the crack (toward the observer); x and y axes above the first snapshot at the upper left of the figure provide orientation as specified in Figure 3. The crack is 20 grid steps long and 20 grid steps wide, and the snapshots are taken every 20 time steps, as indicated at the lower left of the plots. The amplitude of the velocity is normalized to a maximum value of unity in each individual plot to facilitate the viewing of the evolution of the wave field over the crack surface.

al and longitudinal modes are easily recognizable in the motion of the wall depicted in the last 12 snapshots. The shape of the initial disturbance is an impulse propagating over a semicircular wave front with the maximum amplitude along the y axis (see Figure 3 for reference). A partial collapse of the crack wall is seen to precede the passage of this wave front in the first six snapshots; this collapse is associated with the shear wave radiated by the rupture and is well illustrated in the displacement plots of Figure 5. The initial motion at the barrier location (grid (1, 1) in the convention of Figure 3) is an opening, but the onset of displacement is a closure at increasing distance from the barrier. This closure of the crack coincides with the shear arrival and is well marked along most profiles (see, for example, the arrival at $\alpha t/L = \sqrt{3}$ at $x = 0$, $y = L$, i.e., grid (1, 20) in profile 1). The reflection from the edge of the crack at $x = W/2$, which is marked by the first vertical arrow in the first three traces depicted at the upper left, appears as a weak wavelet immediately following the impulsive response of the wall on the first six profiles. The reflection from the end of the crack at $y = L$, marked by the second vertical arrow, is easy to follow on these profiles but becomes increasingly weaker on profiles 7–10. Rayleigh wave motion dominates the dynamic response after the first reflection. The elastic energy due to the rupture is rapidly lost by radiation into the solid so that the motion of the crack appears fairly complete at $\alpha t/L = 6$. The initial opening of the crack shows a strong overshoot in the vicinity of the barrier, the amplitude and period of which is related to the size of the barrier. My calculations indicate that the amplitude of the overshoot decreases as the width of the barrier is increased in the x direction, and in the limiting case of a two-dimensional crack suddenly expanding by Δl, this overshoot vanishes [see *Chouet and Julian*, 1985, Figure 1].

A solution for the dynamics of a wet crack, obtained by solving the 12 equations (equations (1), (2), and (3)) is presented in Figures 6 and 7. Figure 6 shows snapshots of the normal component of the particle velocity on the crack surface in the same manner as Figure 4 ($\Delta t' = \alpha \Delta t/L = 1/80$), and Figure 7 displays the normal component of the displacement of the crack wall given in dimensionless units of the crack and

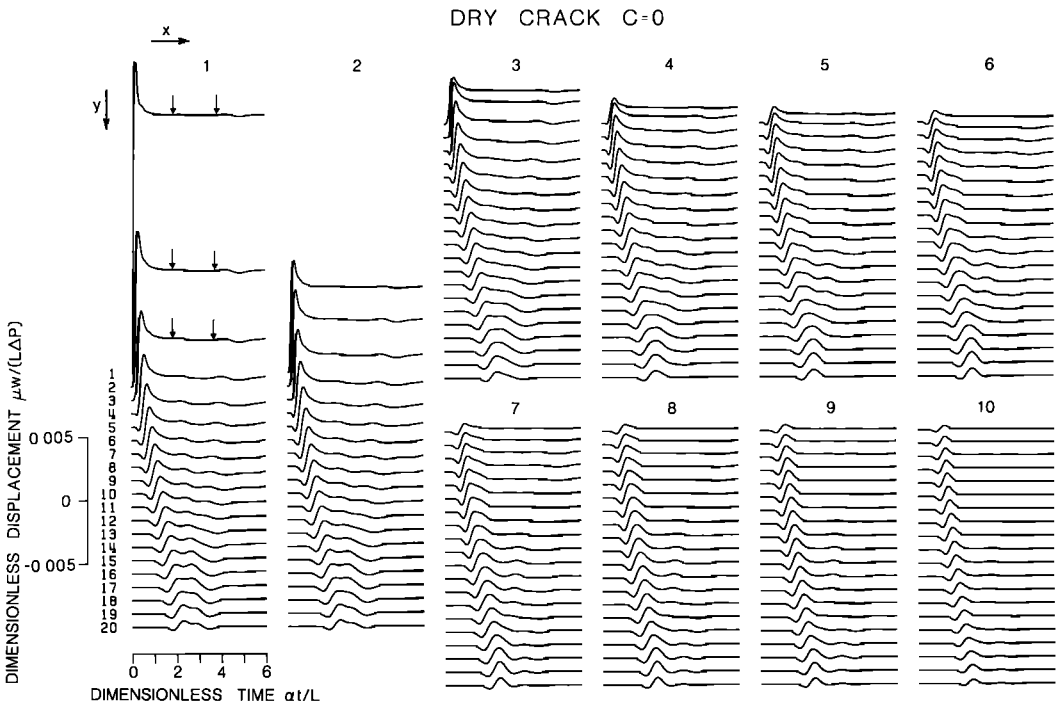

Fig. 5. Normal displacement of the crack wall plotted as a dimensionless function of dimensionless time for a dry crack ($C = 0$) 20 grid steps long and 20 grid steps wide, which expands by $\Delta S = L^2/200$. Due to the symmetry of the problem about $x = 0$, the displacements are shown for the half $x > 0$ of the crack only. The x coordinates of the displacement are indicated at the top of each individual stack of seismograms, and the y coordinates along these profiles are shown at the left of the first stack (note x and y arrows at the upper left of the figure and see Figure 3 for reference). The weak reflections from the edges of the crack, marked by vertical arrows above the first three traces shown at the left, are indicative of the strong loss of energy by elastic radiation into the solid.

medium parameters, as in Figure 5. The fluid in the crack is inviscid ($F = 0$), and the wet crack is characterized by a stiffness $C = 15$ and by the ratios $\alpha/a = 2$, $b/\mu = 0.5$, $W/L = 1$, and $L/d = 30$. The first two ratios imply a ratio between the density of the fluid and the density of the solid given by $\rho_2/\rho_1 = (\alpha/a)^2/(3\mu/b) = 2/3$. This ratio is assumed in most of the calculations presented in this study (see Table 1).

The failure of the barrier produces a strong impulsive opening of the crack at that location. The resulting cavity, however, triggers a drop of pressure in the fluid which causes a partial collapse of the crack wall as this disturbance propagates in the fluid. This negative wave front is clearly seen preceding the impulsive opening of the crack in the first five snapshots displayed in Figure 6. This is again clearly reflected in the profiles shown in Figure 7; on profile 1, for example, the first motion at the barrier location (grid (1, 1) in the convention of Figure 3) is an opening, but this first motion becomes a collapse a short distance away. The collapse of the wall increases in duration as the pressure disturbance spreads across the crack, evolving into a long-period signal with a negative onset followed by a positive recovery which is interrupted by the first lateral reflection from $|x| = W/2$. This simple picture observed in the vicinity of the barrier develops into a more complex signal showing first motions with polarities that are first positive, then negative at larger distances. The compressional and shear wave generated in the solid by the rupture are not observable in the wall response away from the barrier. The first well-developed wave front propagating over the length of the crack is that due to the crack wave, which is similar in nature to the tube wave propagating in a fluid-filled borehole [*Biot*, 1952; *White*, 1965]. *Biot* [1952] referred to this wave by the appellation of Stoneley wave by analogy to the Stoneley waves propagating at a liquid-solid interface. The crack wave propagates with a velocity lower than the velocity of sound in the fluid; it is dispersive, and for longer wavelengths it is analogous to the wave studied in the water-hammer phenomenon (see discussion below and in the following section).

The first reflection from the left and right edges of the crack (see snapshots 80–160 in Figure 6) is marked by an arrow at the barrier location (grid (1, 1)) in Figure 7. Contrary to the situation depicted in Figure 5 for the dry crack, this reflection is well developed and easy to follow on each profile all the way to its inception on profile 10. Interference between the two lateral reflections and the wave front reflected from the crack edge at $y = L$ increases the complexity of the pattern of displacement for $\alpha t/L > 3$. The velocity plots shown in Figure 6 show well the development of the interference pattern between the lateral and longitudinal modes of vibration of the crack (see snapshots starting at 240). The data in Figures 6 and 7 cover approximately one full cycle of oscillation over the crack length. The only mechanism of dissipation is elastic radiation into the solid.

The dependence of the impulse response of the fluid-filled crack on the stiffness parameter C is investigated in Figures 8–11. In Figures 8–11 I have kept the ratio b/μ fixed to the value 0.5 and varied the stiffness through the ratio of crack length to crack thickness (see Table 1). The ratio W/L is allowed to vary also but the ratios α/a and ρ_2/ρ_1 remain fixed to their original values. The fluid is inviscid ($F = 0$) as in Figures 6 and 7. Figure 8 shows the solution obtained for the normal component of displacement of the crack surface when the stiffness is $C = 50$, Figures 9 and 10 represent a crack with the

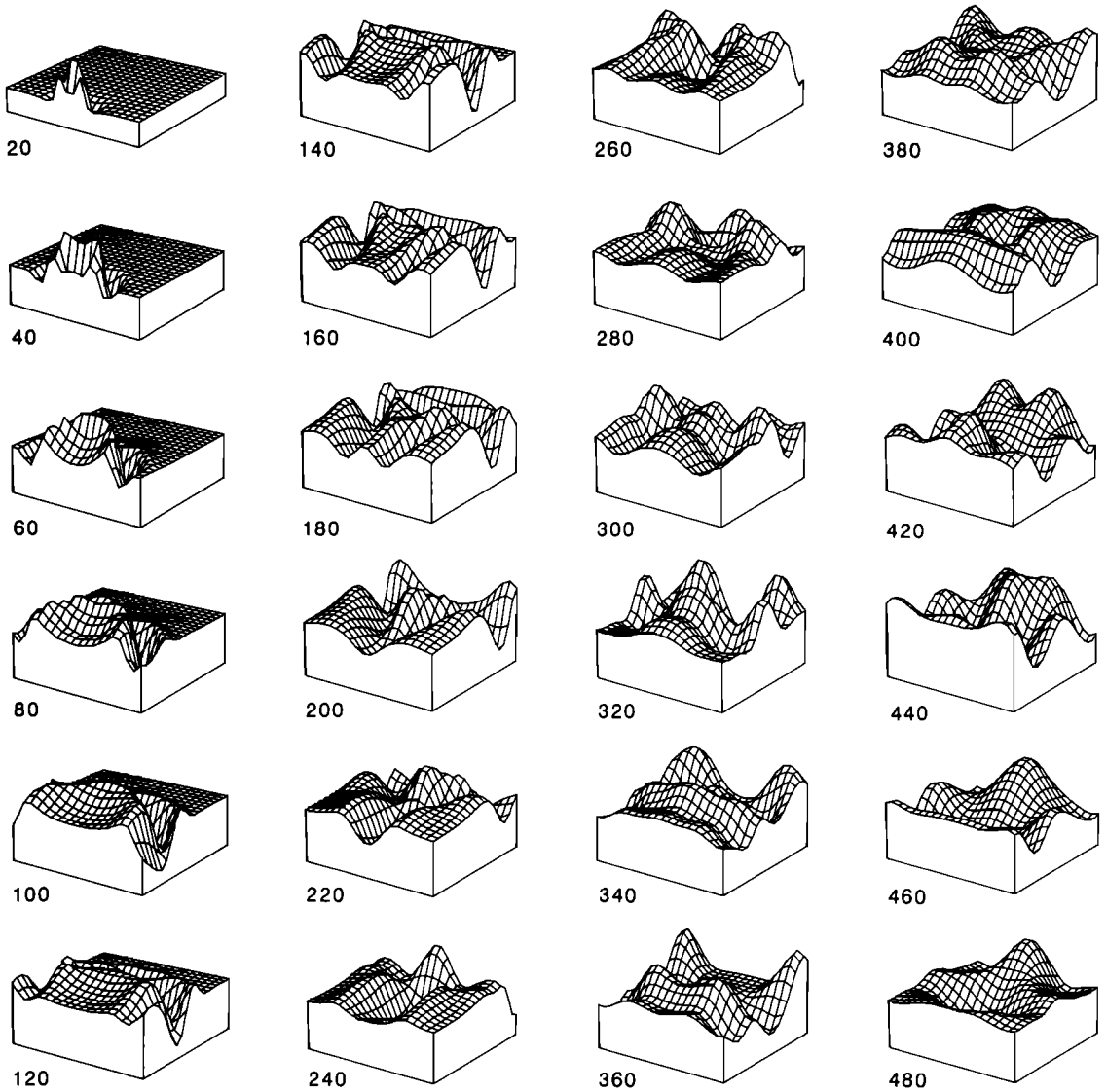

Fig. 6. Snapshots of the normal component of the particle velocity on the surface of a fluid-filled crack with the stiffness $C = 15$. The crack is 20 grid steps long and 20 grid steps wide, and the snapshots are taken every 20 time steps. Amplitudes are normalized as in Figure 4.

stiffness $C = 100$, and Figure 11 shows the solution for a crack with the stiffness $C = 250$. The ratios L/d used in these models range from 100 to 500 and are summarized in Table 1.

The set up of Figure 8 is identical to that of Figure 7 with $\Delta t' = \alpha \Delta t/L = 1/80$. The crack shown in Figures 9 and 10 is 40 grids long and 20 grids wide corresponding to $W/L = 0.5$, so that the rupture area is reduced by a factor of 4 compared to the earlier models. The dimensionless time step for this model has the value $\Delta t' = 1/160$ and the response is obtained up to $\alpha t/L = 3$. Similarly, in Figure 11 the crack is discretized by 100 grids over its length and 20 grids over its width, corresponding to $W/L = 0.2$, so that the rupture area is 25 times smaller than that considered in the original crack model. In that case, only 20 grids representing a section one quarter of the crack length starting at the barrier have been plotted along each profile. The dimensionless time step for this solution has the value $\Delta t' = 1/400$, and the calculation extends to a dimensionless time of 1, showing the details of the onset of the signal near the rupture.

The most obvious feature observed in Figures 8–11 is the decrease of the wave speed of the crack wave as the stiffness increases. For example, the wave speed in a crack with the stiffness 100 (Figure 10) is about 1/2 of the speed in a crack with the stiffness 15 (Figure 7). Also apparent in Figures 8–11 is the dependence of the crack wave velocity on wavelength. The crack wave is dispersive, showing a phase velocity approaching the acoustic velocity of the fluid at short periods and decreasing with longer periods. This dispersion is evident in the snapshots of Figure 6 and 9, where the initial wave front propagating over the length of the crack consists of a short-period wavelet followed by a longer-period component. The dispersion is also clearly seen at the location of grid (1, 20) in Figures 8, 10, and 11, where the longer-period dominant phase of the signal has a shorter-period precursor. The dispersion characteristics of the crack wave and the dependence of its wave speed on the crack stiffness are similar to the behavior of the Stoneley phase in a fluid-filled pipe [see *Biot*, 1952, Figure 8]. A quantitative analysis of these dispersion characteristics as a function of wavelength and stiffness can be performed from measurements of the spectral peaks of resonance of the

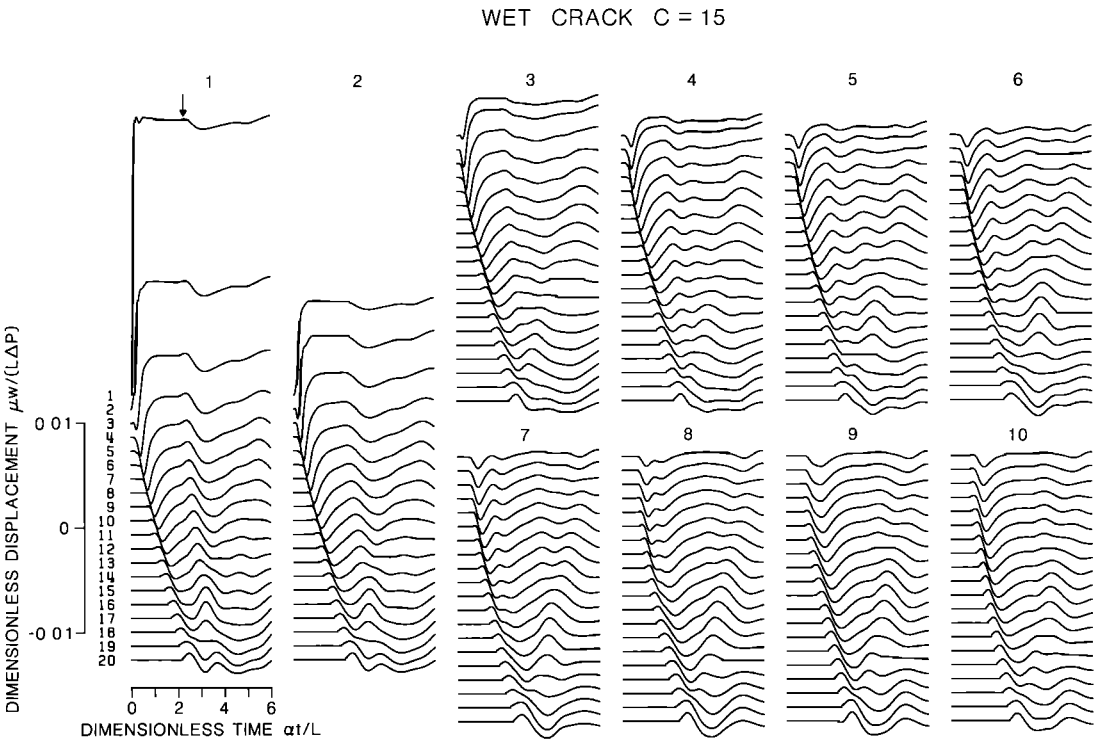

Fig. 7. Normal displacement of the crack surface in dimensionless units for a fluid-filled crack 20 grid steps long and 20 grid steps wide, which expands by $\Delta S = L^2/200$. The x and y coordinates of the displacement are indicated at the top and left, respectively (see Figures 3 and 5 for reference). The crack has the stiffness $C = 15$.

source in the frequency domain. This question will be addressed in the next section.

Another significant effect of the crack stiffness is the predominance of longer periods of motion in the response of stiffer cracks (compare, for example, Figures 7 and 8). Finally, the small overshoot associated with the rupture (see Figure 7) disappears as the crack stiffness increases. In Figures 8, 10, and 11 this overshoot is replaced by a gradual rise of the displacement to its static level.

Figure 9 shows the normal component of particle velocity on the crack surface for a crack with the stiffness $C = 100$. The disposition of the snapshots is the same as in Figure 6, but the

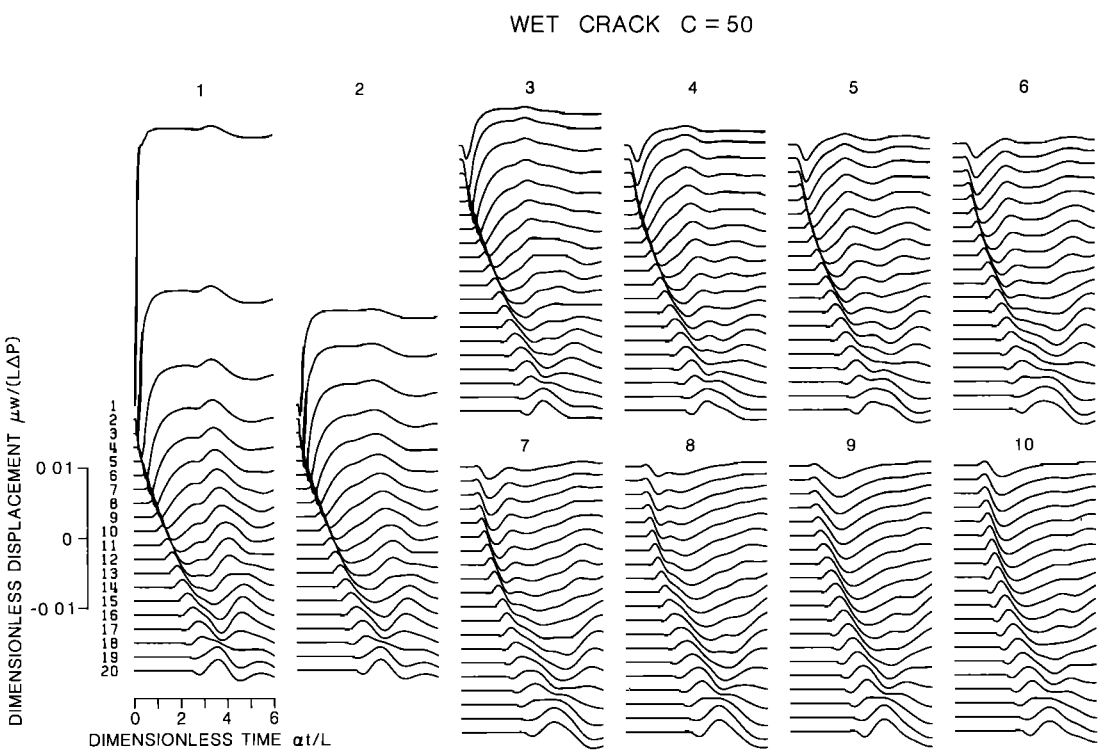

Fig. 8. Same as Figure 7 for a crack with the stiffness $C = 50$.

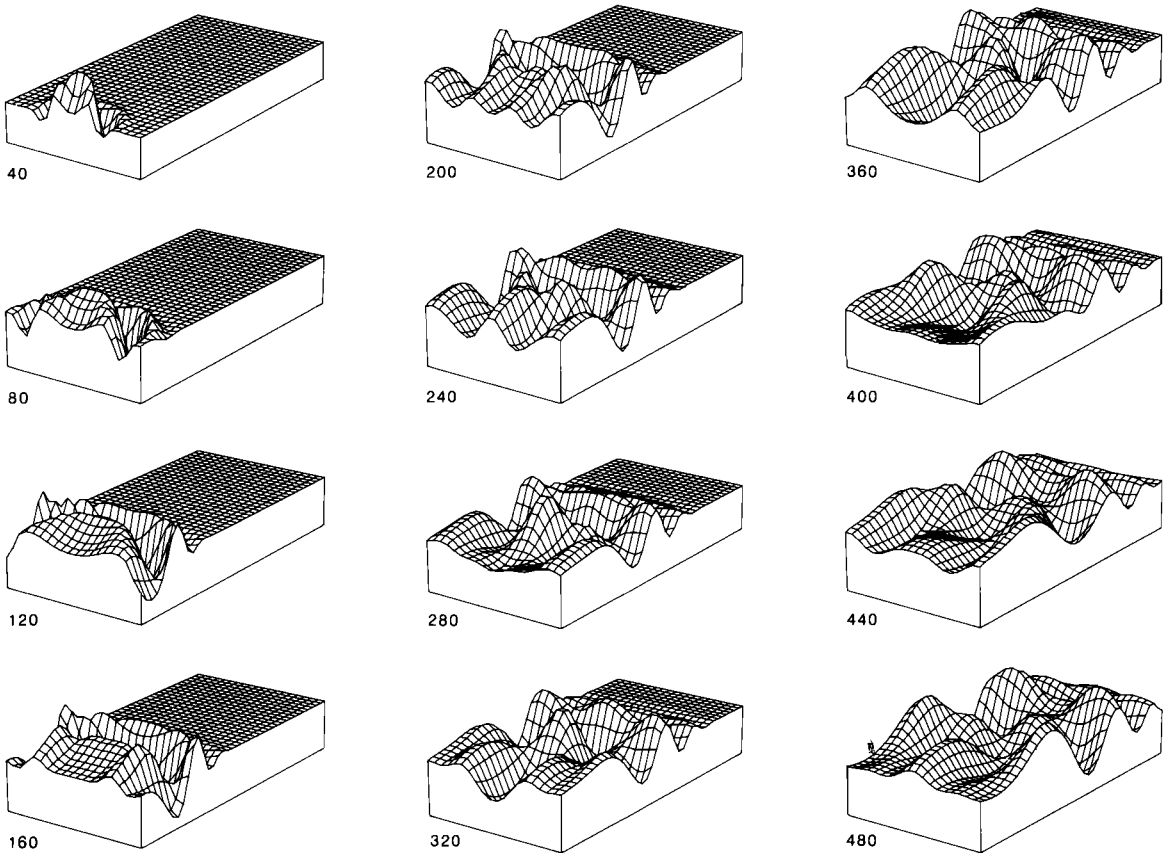

Fig. 9. Snapshots of the normal component of the particle velocity on the surface of a crack with the stiffness $C = 100$. The crack is 40 grid steps long and 20 grid steps wide ($W/L = 0.5$), and the snapshots are taken every 40 time steps. The amplitudes are normalized as in Figure 4.

crack is 40 grids long and the snapshots are taken at intervals of 40 time steps in the present case (the time step has the value $\Delta t' = 1/160$). Since the crack is twice as long as it is wide, the onset of the first two lateral reflections near $80\Delta t'$ ($t' \simeq 0.5$) and $400\Delta t'$ ($t' \simeq 2.5$) is not obscured by the first longitudinal reflection occuring near $480\Delta t'$ ($t' \simeq 3$). The displacements of the crack wall (Figure 10) show further details of the pattern of the wall response due to these reflections. The crack being narrower, the first lateral reflection occurs earlier than in the previous models. This reflection appears as the well-marked secondary wavelet on profiles 1–8. The second lateral reflection is also easy to spot as a third wavelet on profiles 5–7. The merging of the first lateral reflection into the initial wave front propagating over the length of the crack broadens the dominant wave associated with the first arrival (for example, observe the period of the dominant wave between grids 15 and 25 in profiles 4–8). While the first motion consists of a short-period negative pulse in the vicinity of the rupture, the period of the motion associated with the collapse of the crack wall becomes distinctly longer as the distance from the barrier increases. This period is the longest near the lateral edges of the crack (see profiles 8, 9, and 10). Note also that the frequency content of the first motion near the rupture location increases as the rupture area decreases. This is apparent when comparing the period of the negative onset near the rupture in Figures 7 and 10.

The model shown in Figure 11 depicts a crack with the stiffness $C = 250$. As mentioned above, the crack is 100 grids long, but the displacements are obtained only over the first 20 grids, showing the details of the motion in the vicinity of the rupture. Although the frequency content of the crack response generally decreases with increasing stiffness, in this case the frequency content of the signal is higher than in the previous model because the rupture area is smaller. The first reflection also occurs earlier as a result of the smaller ratio of crack width to crack length ($W/L = 0.2$).

The effect of the viscous dissipation in the fluid is illustrated in Figure 12, which compares the crack response obtained using the viscous damping parameter $F = 0.1$ (Figure 12*a*) with that obtained earlier for an inviscid fluid (Figure 12*b*). The crack stiffness is $C = 100$, and the data cover roughly one full cycle of flow oscillation over the crack length. Note that if one assumes a crack thickness of 0.5 m, length of 100 m implied by the stiffness, fluid density of 2500 kg/m^3, and compressional velocity of 3000 m/s, the value $F = 0.1$ implies a viscosity of the fluid of 2500 P. These parameters are typical of a shallow Hawaiian dike [*MacDonald*, 1972; *Swanson et al.*, 1976; *Chouet*, 1981]. The viscous shear stress applied by the fluid at the crack wall results in a net opening that is 30% larger at the barrier location and about 40% smaller at the opposite end of the crack (compare the amplitudes along profile 1 in Figures 12*a* and 12*b*). There is also a marked overshoot of the signal in the vicinity of the barrier, which is absent for an inviscid fluid. The energy loss due to viscous dissipation at the wall reduces the amplitude of the reflections from the edges of the crack and strongly attenuates the high-frequency content of the crack resonance. The net result is a shorter duration of the dynamic motion. This point will be further emphasized in the section treating the far-field radiation from the source. Other than the features discussed

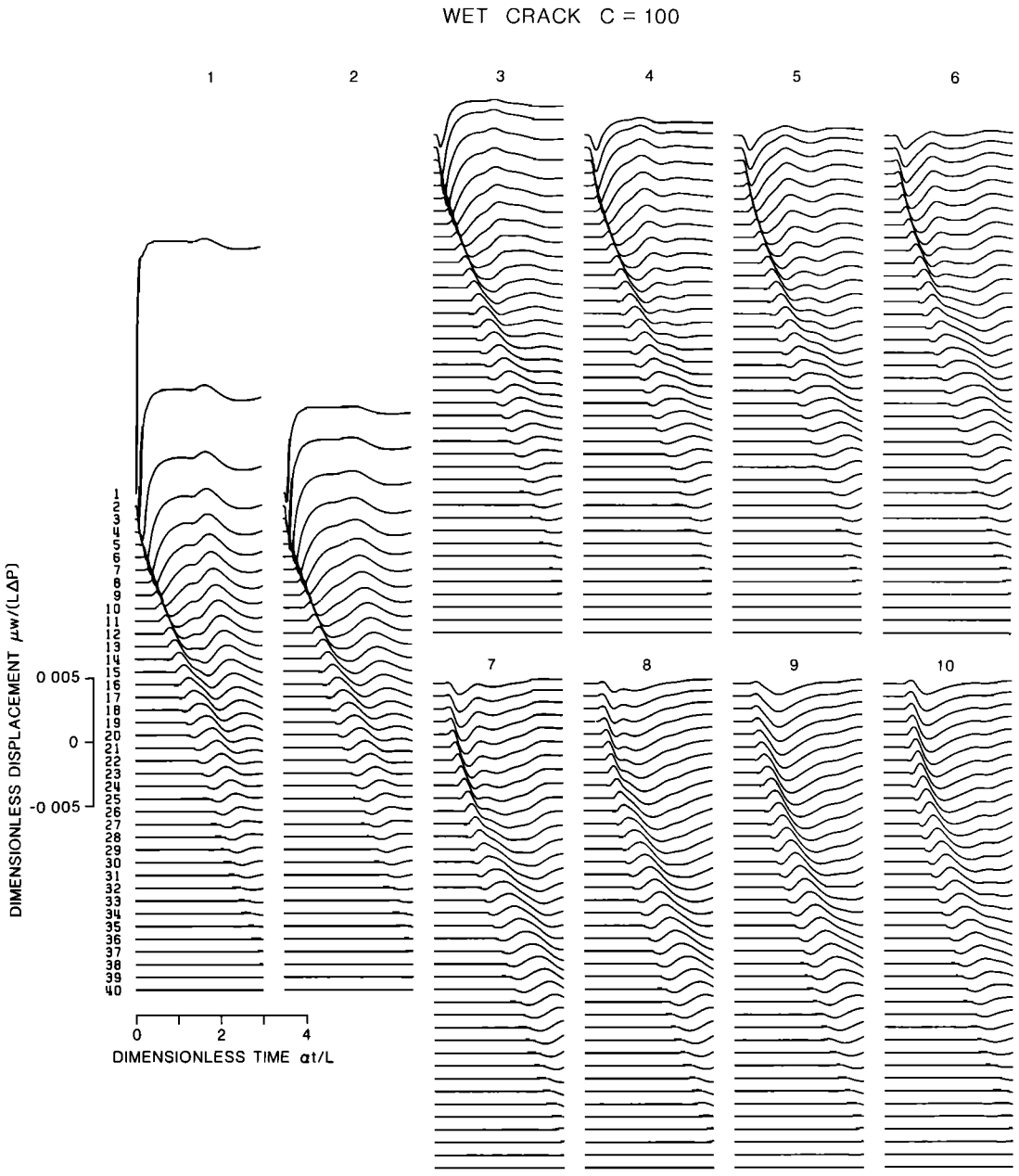

Fig. 10. Dimensionless normal displacement of the crack surface for a crack 40 grid steps long and 20 grid steps wide ($W/L = 0.5$) which expands by $\Delta S = L^2/800$. The disposition of the profiles is the same as in Figure 7, and the crack has the stiffness $C = 100$.

above, the characteristics of the wall response for a viscous fluid are similar to those obtained for an inviscid fluid.

CRACK RESONANCE AND WAVE SPEED OF THE CRACK WAVE

To gain a quantitative understanding of the excitation of the various modes of resonance of the source, it is necessary to carry out the simulation to longer time windows than those considered so far. This analysis is conducted using the transparent boundary conditions discussed earlier; the results are presented in Figures 13–16, and the pertinent crack parameters are listed in Table 1. For a crack containing an inviscid fluid ($F = 0$) the only mechanism of dissipation is elastic wave radiation into the solid. The effect of this loss is obvious in Figure 13, which displays the resonance of the crack sampled along the first profile ($x = 1$) of Figure 10. The crack parameters are identical to those of Figure 10. The data displayed represent about six cycles of flow oscillation over the crack length. High frequencies are quickly dissipated over the first few cycles of oscillation. Low frequencies, on the other hand, show a much weaker damping, indicating that the radiated elastic energy is a small part of the available acoustic energy in the fluid at those frequencies. The resulting signal has a characteristic signature, which is enriched in high frequencies near its onset and becomes enriched in low frequencies over the duration of resonance. A similar result was obtained by *Chouet and Julian* [1985] in their study of the dynamics of a fluid-driven crack in two dimensions. Although this behavior seems at odd with the suggestion made by *Aki et al.* [1977] that high-frequency oscillations may be trapped in the near field of the source, it must be emphasized that the idealized

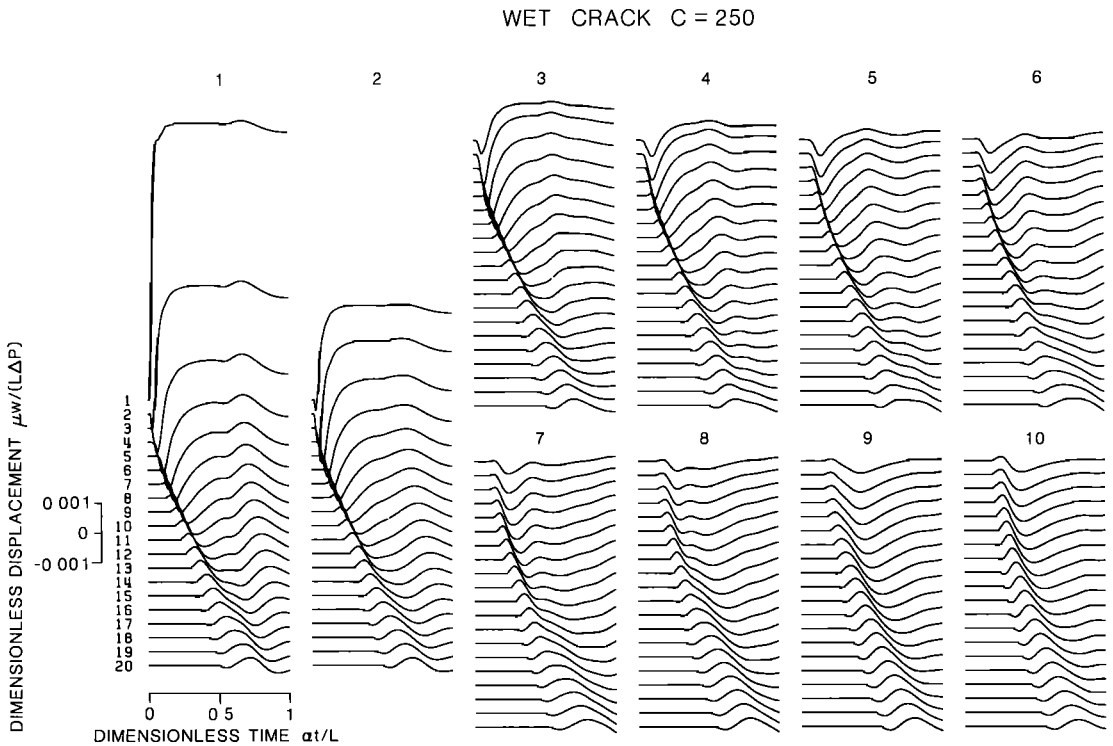

Fig. 11. Dimensionless normal displacement of the crack wall for a crack 100 grid intervals long and 20 grid intervals wide ($W/L = 0.2$) which expands by $\Delta S = L^2/5000$. The disposition of the profiles is the same as in Figure 7, but only the first 20 grids in the y direction are shown. The crack has the stiffness $C = 250$.

crack model used in the present study cannot fully resolve the finer scale of the complex interactions between the liquid and the solid. For example, *Chouet and Julian* [1985] described the existence of pseudo-Rayleigh waves in one of their more refined models. A detailed analysis of wave propagation in a fluid-filled borehole made by *Cheng and Toksöz* [1981] also demonstrated the complexity of the wave train due to the interaction between the two media. An improved model that includes the details of the viscoelastic field within the fluid will be required to provide further insight into the effects of crack geometry and fluid properties on the high-frequency characteristics of the wave field near the source.

The spectral characteristics of the crack resonance are illustrated in Figure 14 for the two-dimensional model of an expanding wet crack of *Chouet and Julian* [1985]. The crack is set in the plane $y = 0$ and extends from the origin 0 to the distance L along the x axis and to $\pm\infty$ in the z direction; the crack expansion occurs at the tip located at $x = L$. The x and y components of velocity are denoted by $\dot{u}$ and $\dot{v}$, respectively; the longitudinal component shown at the top is obtained on the plane $y = 0$, but the transverse component shown at the bottom is obtained half a grid space away from that plane because of the staggered grid used in the finite difference scheme (in this solution the crack was discretized by 40 grids, so half a grid space represents the distance $L/80$). The calculation of the source response was performed over a window extending to a dimensionless time of 100, including about 20 cycles of flow oscillation. The dimensionless frequency band extends from 0 to 1.5, and the spectra are obtained over the range $-L/2 \leq x \leq +3L/2$. Figure 14 depicts very clearly the first seven modes of resonance of the source; the dominant mode corresponds to the third overtone on both components of motion. The fundamental mode for the longitudinal component of motion has the wavelength $2L$, while that for the transverse component has a wavelength L. The latter shorter wavelength reflects a change of sign in the polarity of the normal component of motion, which is related to the characteristics of the pressure field within the crack (the initial opening of the crack tip produces a pressure drop in the fluid, which results in a partial collapse of the crack wall away from that tip). This change of sign accounts for an extra node in each individual mode displayed by the transverse component of motion, which is absent in the longitudinal component.

The resonance of the three-dimensional crack has characteristics similar to those displayed in Figure 14, with the added complexity due to the lateral modes of oscillation. An example of the crack excitation triggered by the failure of the barrier is given in Figure 15 for a crack whose parameters are identical to those of Figure 10. Figure 15 is a contour map of the spectral amplitude of the normal component of particle velocity on the crack surface given as a function of the dimensionless frequency $\nu L/\alpha$ and distance along a profile parallel to the y axis; the profile depicted in Figure 15 is positioned at $x = 5$ (see Figure 3 for orientation). Position along the profile is indicated by grid numbers; the bottom edge of the plot is located at grid (5, 1), at a distance $W/4$ from the barrier. The spectra are representative of the complete history of crack excitation calculated over a dimensionless time window extending to $\alpha t/L = 100$. The ratio of crack width to crack length, $W/L = 0.5$, and the location of the barrier at the middle of the crack edge $|x| \leq W/2$, $y = 0$ are such that the period of the fundamental longitudinal mode is 4 times that of the fundamental mode in the lateral direction. The interaction between the lateral and longitudinal modes of oscillation results in a progressive degradation in the spectral alignment of the resonant peaks along the profile at frequencies above the

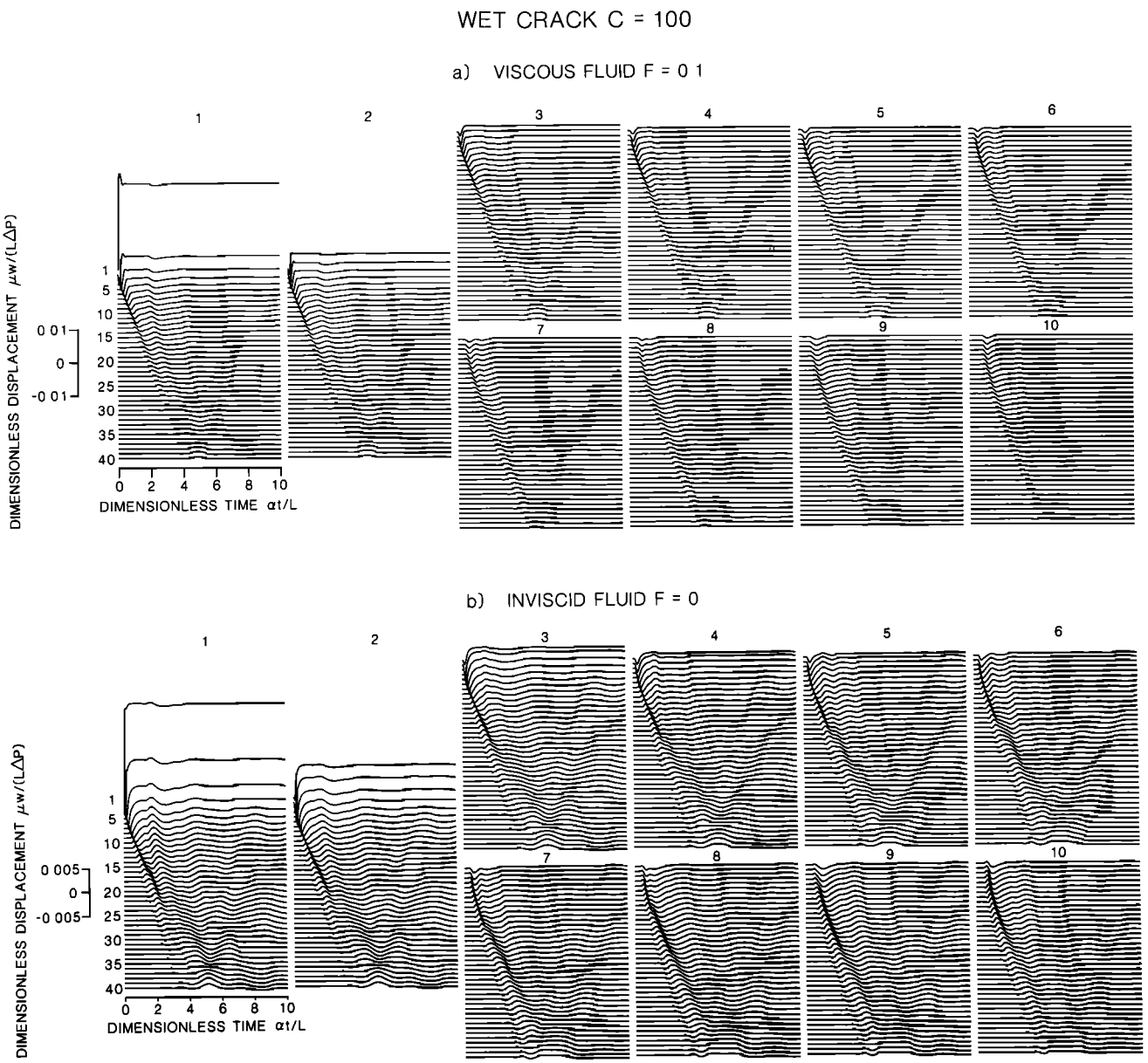

Fig. 12. Effect of fluid viscosity on the crack response. (*a*) Dimensionless normal displacement of the crack surface for a crack with the stiffness $C = 100$ containing a viscous fluid characterized by the parameter $F = 0.1$ (see text for details). The crack is 40 grid steps long and 20 grid steps wide ($W/L = 0.5$) and expands by $\Delta S = L^2/800$. The disposition of the profiles is the same as in Figure 10. (*b*) Response of the crack depicted in Figure 12*a* when the fluid is inviscid. The crack parameters are identical to those used in Figure 10.

third overtone. This is marked by a shift in the frequency of the peaks near grid 2 in the band $0.35 \leq vL/\alpha \leq 0.55$, followed by the gradual disappearance of individual peaks at frequencies $vL/\alpha > 0.55$.

Figure 16 shows samples of spectra of the normal component of particle velocity obtained at selected locations on the crack surface. As in Figure 15, the spectra are representative of the complete crack response calculated over a dimensionless time window extending to $\alpha t/L = 100$. The x and y coordinates at which the spectra are obtained are indicated at the upper right of each plot, along with the dimensionless amplitude of the dominant mode of resonance. The spectra are normalized to the amplitude of this dominant mode. The first 13 longitudinal modes of resonance are indicated above the upper two plots. Note that grid (5, 40) and grid (5, 21) correspond to two horizontal profiles drawn along the top edge and midsection of the contour map displayed in Figure 15. Only the even modes are seen at grids (5, 21) and (3, 21) because these positions correspond to nodes for the fundamental mode and odd overtones (compare grid (5, 21) with data of Figure 15). The fundamental lateral mode appears as a secondary peak close to the third longitudinal overtone on grid (3, 40) and grid (3, 21). Modes above the twelfth longitudinal overtone are not apparent, indicating that there is little or no energy in the higher modes of the source. This characteristic was observed in all my models and is consistent with the synthetics displayed in Figure 13, which show that high frequencies are quickly dissipated by elastic radiation so that there is no significant energy available for the excitation of the source resonance at those frequencies.

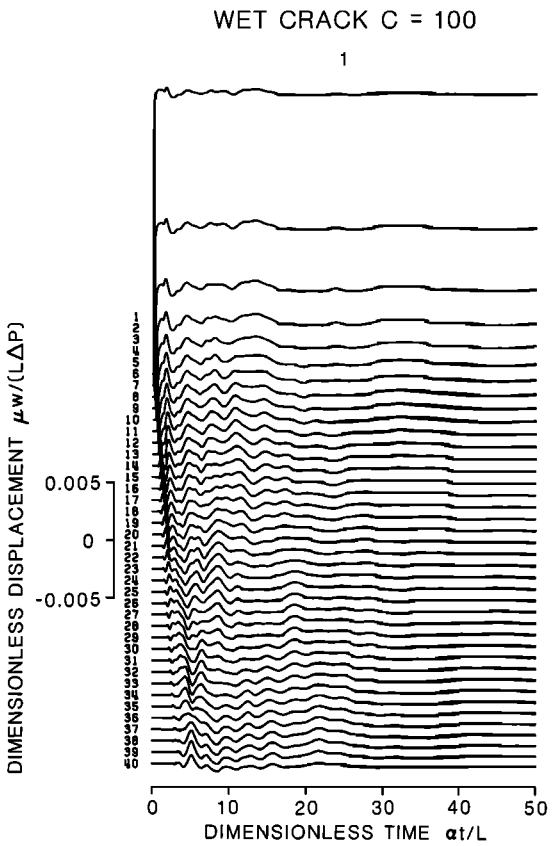

Fig. 13. Dimensionless normal displacement of the crack surface depicted along the first profile of Figure 10 and showing the details of the crack resonance triggered by the failure of the barrier.

The frequencies and wavelengths of individual modes are easily measured from data such as those displayed in Figures 15 and 16. The frequencies of the first 13 longitudinal modes obtained in that manner are listed in Table 2 for various crack stiffnesses ranging between 5 and 100. Using the ratio $\alpha/a = 2$ and the formula $\lambda = \mathbf{v}L/(v'\alpha)$, in which $v' = vL/\alpha$ denotes the dimensionless frequency of a mode, one obtains the relation $\mathbf{v}/a = 2v'(\lambda/L)$. A plot of the dispersion curves $\mathbf{v}/a$ versus the dimensionless wavelength λ/L is given in Figure 17 for the crack stiffnesses listed in Table 2. These curves show the same qualitative behavior as those obtained by *Biot* [1952] for the propagation of waves in a fluid-filled bore. As noted earlier, the crack wave propagates with a velocity lower than the velocity of sound in the fluid and shows a strong dependence on wavelength and crack stiffness. For a crack stiffness of 100 the phase velocity of the crack wave is in the range $0.1a < \mathbf{v} < 0.4a$ for wavelengths in the range $0.1L < \lambda < 1.0L$. A plot of the ratio $\mathbf{v}/a$ as a function of stiffness C is given in Figure 18 for the individual modes of resonance of the crack. This plot shows that the phase velocity of the crack wave tends toward a horizontal asymptote for large stiffnesses. That asymptote corresponds to $\mathbf{v}/a \sim 0.1$ for the fundamental mode. As the ratio b/μ is fixed to 0.5 in my calculations, the stiffness range used in Figures 17 and 18 and Table 2 is equivalent to a range of aspect ratios L/d between 10 and 200. Since both L/d and b/μ similarly affect the crack stiffness, the behavior of the crack wave at $\lambda/L \sim 1$ is analogous to that of the wave studied in the theory of the water-hammer phenomenon [see *Biot*, 1952, equation 3.15]. For short wavelengths, one would expect that the velocity of the crack wave should coincide with the Stoneley wave velocity at a plane interface between two semi-infinite liquid and solid media. As seen above, however, this portion of the dispersion curve cannot be measured because there is no significant excitation of the higher modes of the source.

FAR-FIELD COMPRESSIONAL WAVE

Using the representation theorem [*Aki and Richards*, 1980, equation 2.41], the compressional wave radiated by the crack can be computed directly from the opening-velocity function derived earlier. In doing so, I assume that the crack has no initial thickness and that the traction is continuous across the crack. These assumptions allow me to drop one of two terms from the representation theorem and describe the tensile crack by an equation similar to that of shear faulting. The same approach has been used in previous studies of the radiation from fluid-filled cracks [*Aki et al.*, 1977; *Chouet*, 1981; *Chouet and Julian*, 1985], where it was assumed that the finite preexisting thickness of the crack had a negligible effect on the radiated waveforms for wavelengths $\lambda \gg d$. The assumption of continuity of traction across the crack naturally follows from the neglect of crack thickness when the fluid is inviscid. For a viscous liquid, however, the drag applied by the fluid at the crack wall introduces a discontinuity in shear stresses across the crack, and an additional term enters into the far-field equation. The contribution of this additional force to the far-field radiation will be investigated in more details later in this section. For the moment I shall consider the radiation properties of a crack containing an inviscid fluid under the assumptions stated above. Referring to the coordinate system shown in Figure 19, I write the radial displacement in the far field, u^R, in the form

$$u^R(\mathbf{R}, t) = \frac{1}{4\pi\rho\alpha^3 R} S_{\theta\phi} \int_{-w/2}^{+w/2} dx \int_0^L dz \cdot \Delta\dot{v}\left(x, z, t - \frac{R - x \sin\theta \cos\phi - z \cos\theta}{\alpha}\right) \quad (15)$$

where $\mathbf{R}$ is the position vector of the observer, $R = |\mathbf{R}|$, x and z are the coordinates of a point on the crack surface, α is the P wave velocity, ρ is the density of the medium, $S_{\theta\phi} = \lambda + 2\mu \sin^2\theta \sin^2\phi$ is the radiation pattern of the compressional wave [e.g., *Chouet*, 1979], θ and ϕ are the angles in the spherical coordinates, and $\Delta\dot{v}$ is the source opening-velocity function.

Rather than computing the far-field radiation directly from (15) by numerical integration, a difficult task due to singularities in $\Delta\dot{v}$, a more convenient approach is to work in the frequency domain. Assuming $\lambda = \mu$, $\alpha = \sqrt{3}\beta$, the Fourier transform of (15) is obtained in the form

$$u^R(\mathbf{R}, \omega) = \frac{1}{12\pi\alpha R}(1 + 2\sin^2\theta \sin^2\phi) \int_{-w/2}^{+w/2} dx \cdot \int_0^L dz\, \Delta\dot{v}(x, z, \omega) \exp\left[\frac{i\omega}{\alpha}(x \sin\theta \cos\phi + z \cos\theta)\right] \quad (16)$$

where the propagation delay $\exp(-i\omega R/\alpha)$ has been removed for convenience. High-frequency components that are contaminated by numerical noise are suppressed by convolving $u^R(\mathbf{R}, \omega)$ with a three-pole low-pass Bessel filter having a dimensionless corner frequency $v' = 4/\pi$. Once filtered, the result is inverted back into the time domain using a numerical fast Fourier transform.

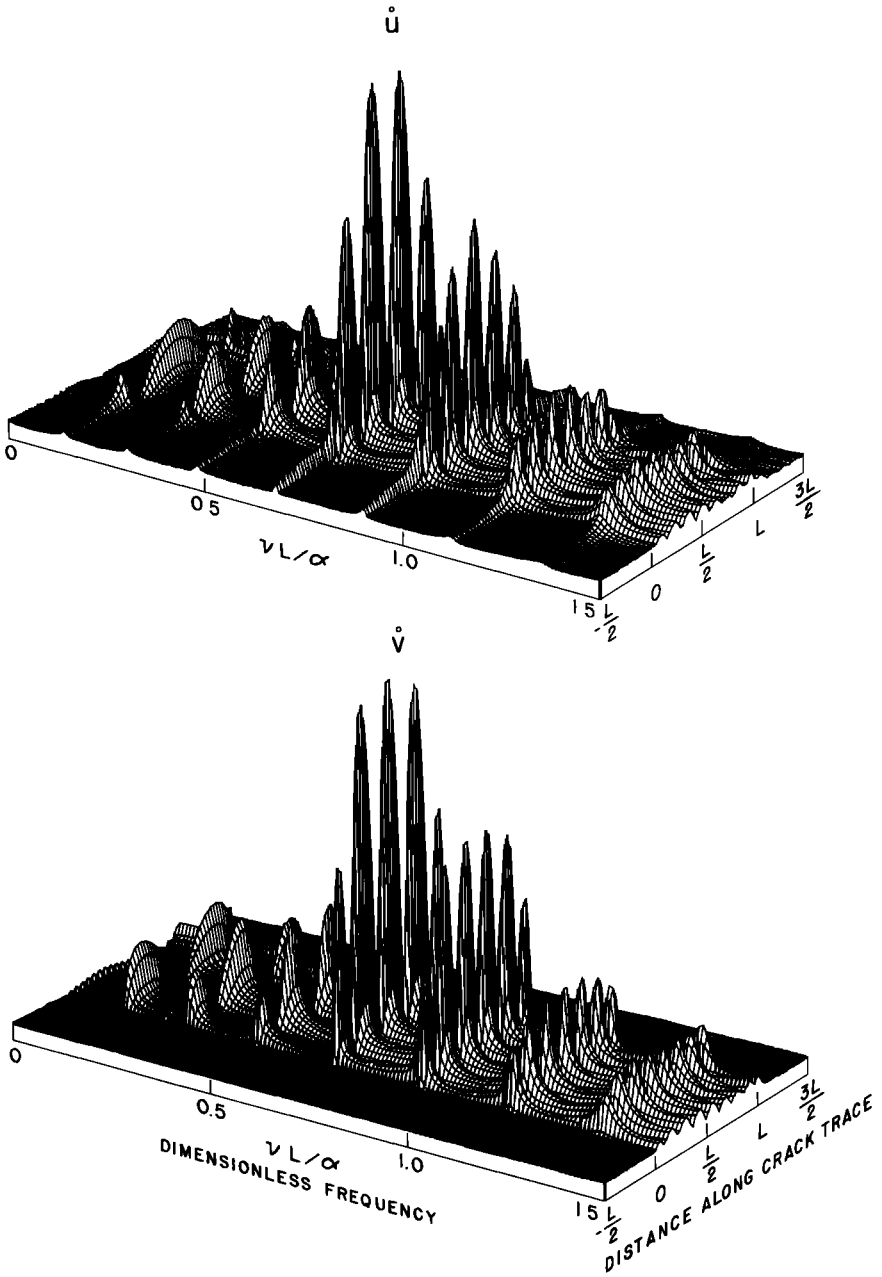

Fig. 14. Spectra of the two components of particle velocity on the crack surface given as a function of the dimensionless frequency $\nu L/\alpha$ and distance along the crack for a two-dimensional crack suddenly expanding by Δl. The components are indicated at the top of each plot: $\dot{u}$, longitudinal; and $\dot{v}$, transverse. The crack has the stiffness $C = 5$ with the parameters $b/\mu = 0.1$ and $L/d = 50$, and the fluid is inviscid.

The computed waveforms are shown in Figures 20–22 for cracks with various stiffnesses. Each figure displays the far-field compressional wave obtained for several angles θ between 0 and 180° and ϕ ranging from 0 to 90°. Because the crack has symmetry about the planes $x = 0$ and $y = 0$, these ranges actually cover the entire sphere of radiation. The results are given as a dimensionless combination of the applied pressure step, the crack length, the distance to the observer, and the rigidity and compressional velocity of the solid.

The far-field radiation varies significantly with the azimuth and elevation under which the crack is seen. In Figure 20, which represents a wet crack with the stiffness $C = 15$, the initial compressive pulse produced by the rupture has a shorter duration at $\theta \simeq 0$ than for larger elevations. For angles θ smaller than 60° this compressive pulse is immediately followed by a small dilatation. For angles θ above 60° the dilatation vanishes, and the compressional first motion lasts up to 5 times as long (compare, for example, the pulse shape obtained at $\theta = 150°$, $\phi = 60°$ with the pulse shape obtained at $\theta = 0°$, $\phi = 90°$). The distinct wave train following the initial compression is due to the reflections off the edges of the crack at $|x| = W/2$. Increasing the stiffness of the crack, but keeping the same rupture area and width to length ratio W/L has the effect of decreasing the amplitude and frequency content of the signal (see Figures 20 and 21). A greater stiffness increases the width of the initial compressive pulse and enhances the amplitude of the signal due to the reflections at the crack edges in relation to this compression for angles θ between 60° and 90°.

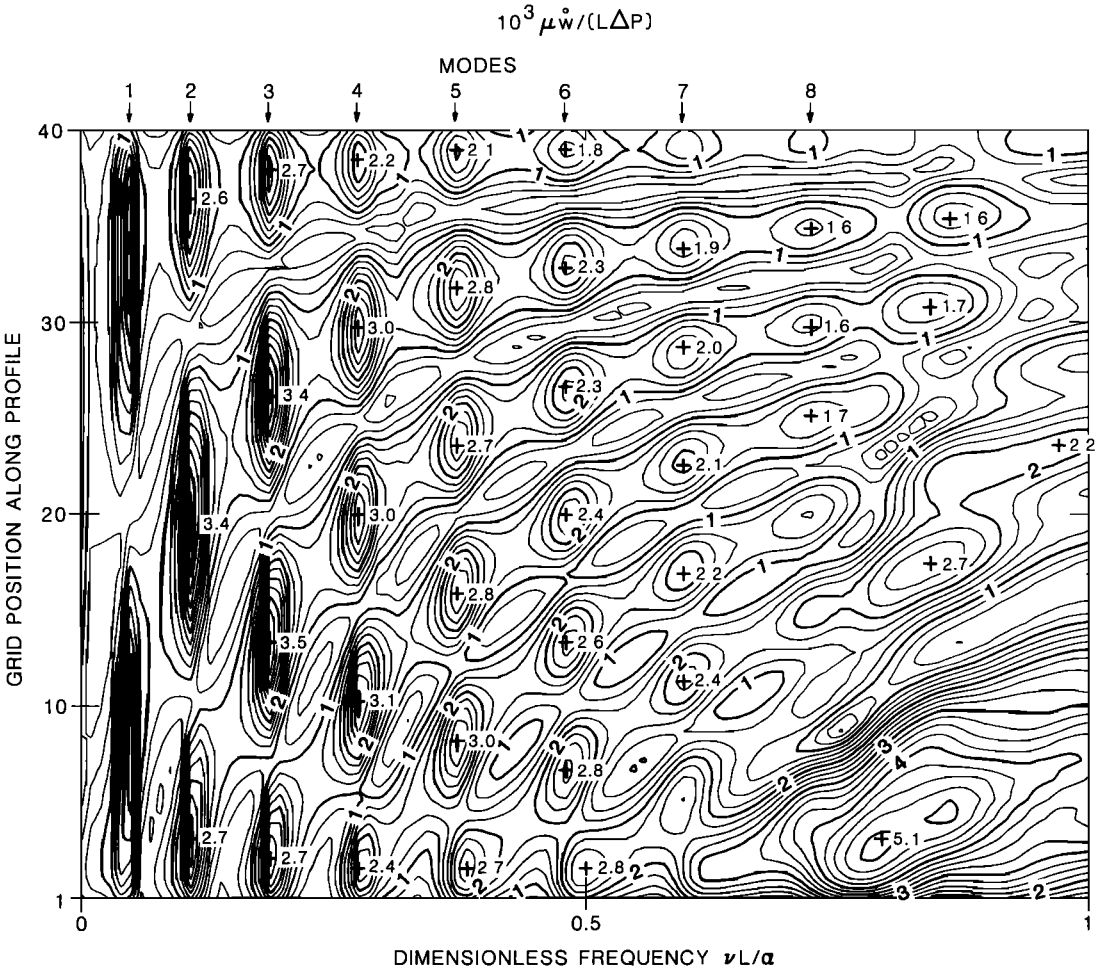

Fig. 15. Contour map of the spectral amplitude of the normal component of particle velocity on the crack surface given as a function of the dimensionless frequency $\nu L/\alpha$ and distance along a profile parallel to the main axis of the crack. The spectral amplitudes are given in dimensionless units as indicated at the top. The frequencies of the first eight longitudinal modes of resonance are indicated by arrows above the contour map.

The small dilatation, which was observed following the initial compression at low angles θ in Figure 20, disappears as the stiffness increases (Figures 21 and 22). The pulse shape of the first compression is also significantly affected for the higher crack stiffness; this is particularly conspicuous for angles θ below 120° and ϕ larger than 60° in Figures 21 and 22. The maximum amplitude of the first compression is found at $\theta = 90°$, $\phi = 90°$, in the direction perpendicular to the crack surface, where it ranges from a dimensionless value $(R/L^2)(\mu/\Delta P)u^R = 1.2 \times 10^{-4}$ for a crack stiffness $C = 15$, to 8.4×10^{-5} for a crack with the stiffness $C = 50$. If one assumes a crack length $L = 100$ m (which implies $W = 100$ m and $\Delta S = 50$ m^2), distance $R = 1000$ m and rigidity of the solid $\mu = 3 \times 10^5$ bars, the maximum displacement calculated ranges between 4×10^{-9} m ($C = 15$) and 3×10^{-9} m ($C = 50$) per bar of applied excess pressure ΔP; thus a pressure $\Delta P = 100$ bars would produce peak displacements of 3×10^{-7} to 4×10^{-7} m in those cases.

The far-field radiation for a crack with the stiffness $C = 100$ is shown in Figure 22. The area of the rupture is smaller than that considered above by a factor of 4, and the width to length ratio W/L of the crack is 0.5. The behavior is similar to that observed in the crack with the stiffness of 50, although the duration of the first compression is reduced owing to the smaller size of the barrier. The conspicuous wave observed near the end of the displayed signal for $\theta < 150°$ is due to the lateral reflections from the edges $|x| = W/2$. The maximum displacement recorded at $\theta = 90°$ and $\phi = 90°$ has a dimensionless amplitude of 1.6×10^{-5} (i.e., $u^R = 5 \times 10^{-8}$ m for $L = 100$ m, $R = 1000$ m, $\mu = 3 \times 10^5$ bars, and $\Delta P = 100$ bars).

The effect of crack stiffness on the far-field P wave train generated by the excitation of the crack is shown in Figure 23. These results are obtained at a fixed receiver position given by $\theta = 30°$ and $\phi = 30°$ (see Figure 19). The dimensionless time window considered has a length $\alpha t/L = 100$ and includes the entire history of crack oscillation; the fluid is inviscid so that the energy loss at the source is purely radiative. The main feature of these synthetics resides in their characteristic long-period signature enriched with short-period components near the onset of the signal. The short duration of the high frequencies in the wave train characterizes the rapid depeletion of high-frequency energy at the source (see Figure 13 and related discussion). Another aspect of interest is the enrichment of the signal in low frequencies as the stiffness increases, which is in harmony with the behavior observed in Figures 7 and 8 and with the dependence of the phase velocity of the crack wave on stiffness seen in Figures 17 and 18. It is of interest to note that the initial short-period compression is followed by a weak longer-period component of dilatation, which could not be resolved in the previous three figures because of the short time window considered for those plots (see

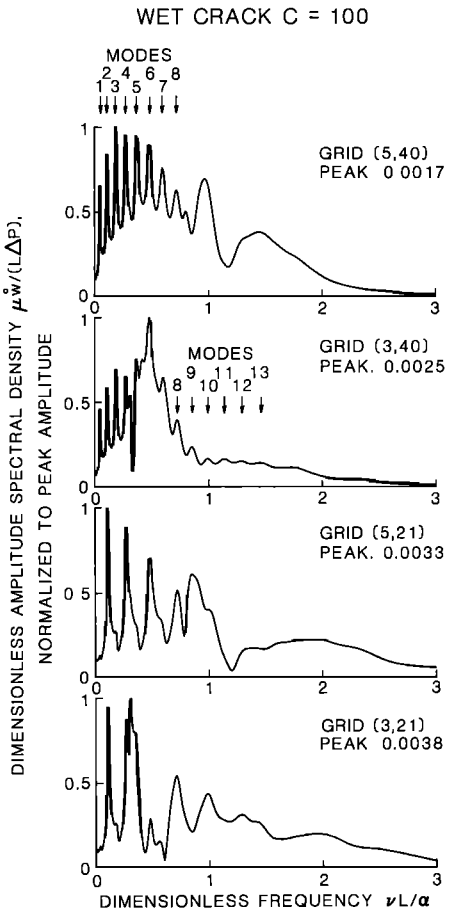

Fig. 16. Spectra of the normal component of particle velocity obtained at selected locations on the crack surface. The crack parameters are identical to those of Figures 10 and 15.

Figures 20–22). This dilatational component is most obvious in the synthetics obtained for a crack stiffness of 100 (bottom trace in Figure 23), where it is associated to a signal with the dimensionless period $T' \sim 20$, which corresponds to the lowest resonant frequency seen in the wave train. The spectra of those waveforms are marked by well-developed narrow-band dominant and subdominant frequencies. The dominant frequency, $\nu' = \nu L/\alpha$, and bandwidth, $\Delta\nu' = \Delta\nu L/\alpha$, of the associated peak have the following dimensionless values for the stiffnesses depicted: $\nu' = 0.78$, $\Delta\nu' = 0.17$ for $C = 5$; $\nu' = 0.40$, $\Delta\nu' = 0.07$ for $C = 50$; and $\nu' = 0.31$, $\Delta\nu' = 0.04$ for $C = 100$. The lowest resonant frequency occurs at $\nu' = 0.15$ for the stiffness of 5, $\nu' = 0.07$ for the stiffness of 50, and $\nu' = 0.05$ for the stiffness of 100.

TABLE 2. Resonant Frequencies for Various Crack Stiffnesses

Mode	Wavelength of Mode λ	Dimensionless Frequency of Mode $\nu L/\alpha$					
		$C = 100$	$C = 75$	$C = 50$	$C = 25$	$C = 15$	$C = 5$
1	L	0.048	0.053	0.068	0.088	0.107	0.160
2	$2L/3$	0.107	0.128	0.157	0.204	0.245	0.341
3	$L/2$	0.185	0.214	0.253	0.329	0.399	0.525
4	$2L/5$	0.273	0.311	0.369	0.476	0.555	0.719
5	$L/3$	0.369	0.419	0.497	0.627	0.729	0.912
6	$2L/7$	0.482	0.544	0.633	0.793	0.910	1.138
7	$L/4$	0.596	0.669	0.776	0.964	1.099	1.349
8	$2L/9$	0.721	0.811	0.934	1.150	1.305	1.562
9	$L/5$	0.852	0.954	1.095	1.336	1.494	1.618
10	$2L/11$	0.991	1.102	1.258	1.517	1.692	···
11	$L/6$	1.138	1.260	1.431	1.713	1.901	···
12	$2L/13$	1.284	1.425	1.607	1.909	2.098	···
13	$L/7$	1.452	1.588	1.807	2.143	2.312	···

When the fluid has a finite viscosity, the traction exerted at the crack wall by the viscous drag of the fluid flowing in the crack results in a net force [*Weertman*, 1971] that needs to be added to the equivalent force system representing the source in the far-field calculation. Referring to Figure 19 and denoting the x and z components of the shear stress applied by the fluid on the crack wall by τ_x and τ_z, respectively, so that the resulting traction discontinuities across the crack face are $\Delta\tau_x$ and $\Delta\tau_z$, one obtains the x and z components of this additional force for an element $dx\,dz$ of crack surface as $\Delta\tau_x(x, z, t)\,dx\,dz$ and $\Delta\tau_z(x, z, t)\,dx\,dz$. The direction of these forces is given by the direction of the flow, so that fluid flowing into the

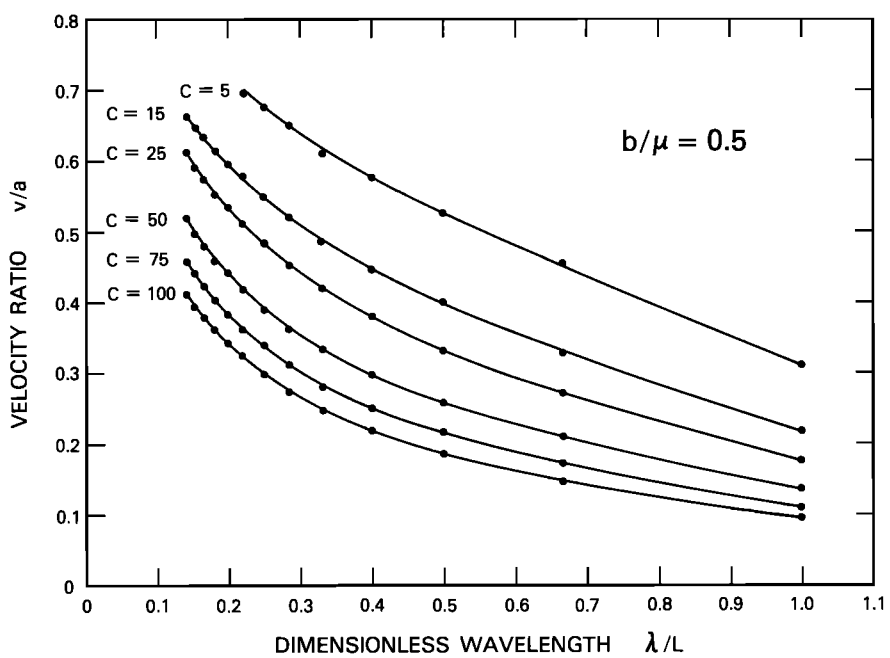

Fig. 17. Phase velocity of crack wave plotted as a function of dimensionless wavelength λ/L for various crack stiffnesses ranging between 5 and 100.

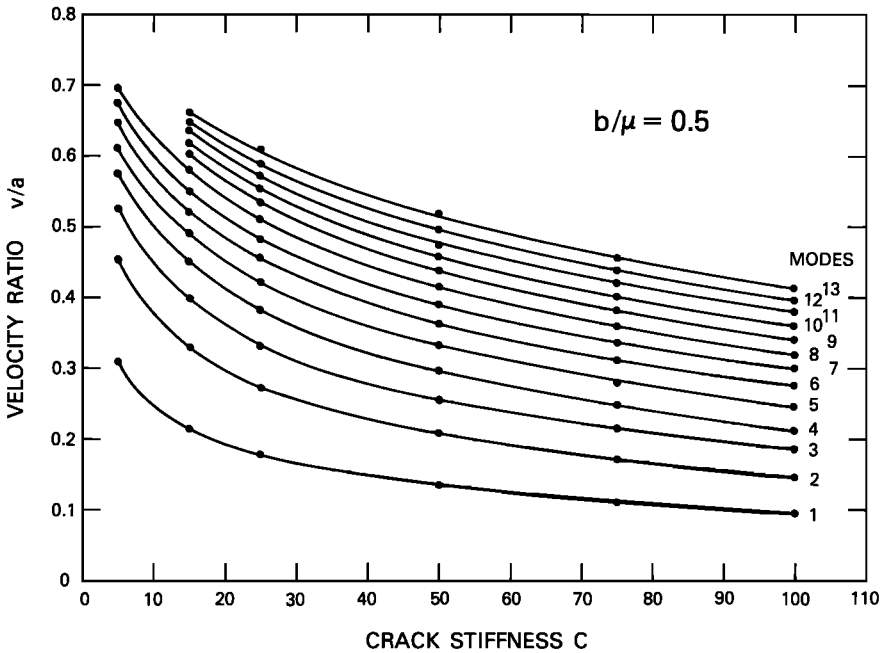

Fig. 18. Phase velocity of crack wave for individual modes of resonance plotted as a function of crack stiffness.

void created by the rupture produces forces directed toward this region. The far-field contribution for each force is a net dilatation in the direction opposite to that of the force and a net compression in the azimuth of the force [see *Aki and Richards*, 1980, Figure 4.2]. Using the coordinate system of Figure 19 and assuming $\alpha = \sqrt{3}\beta$, the Fourier transform of the radial displacement in the far field due to each force component is obtained from the representation theorem [*Aki and Richards*, 1980, equation 2.41] as

$$u^R(\mathbf{R}, \omega) = \frac{1}{12\pi\mu R} \sin\theta \cos\phi \int_{-w/2}^{+w/2} dx \cdot \int_0^L dx\, \Delta\tau_x(x, z, \omega) \exp\left[\frac{i\omega}{\alpha}(x \sin\theta \cos\phi + z \cos\theta)\right] \quad (17)$$

for the contribution due to the force in the x direction, and

$$u^R(\mathbf{R}, \omega) = \frac{1}{12\pi\mu R} \cos\theta \int_{-w/2}^{+w/2} dx \cdot \int_0^L dz\, \Delta\tau_z(x, z, \omega) \exp\left[\frac{i\omega}{\alpha}(x \sin\theta \cos\phi + z \cos\theta)\right] \quad (18)$$

for the contribution due to the force in the z direction. As in (16), the propagation delay over the distance R has been removed from both equations.

To assess the importance of these two terms, I shall consider a crack containing a viscous fluid characterized by the parameter $F = 0.1$. As discussed earlier, this value may be typical for a shallow Hawaiian dike containing a magma with the viscosity of 2500 P. An example of waveforms obtained by summing (16), (17), and (18) is shown in Figure 24, and the individual contributions due to the viscous fluid drag in the x and z directions are displayed in Figures 25 and 26, respectively. Other than the viscosity, the crack parameters are the same as those of Figure 22. The results are given in dimensionless units of the medium and crack parameters and are obtained over a time window that covers the entire history of crack excitation. As in Figures 20–23, the synthetics have been low-pass filtered using a three-pole Bessel filter with a dimensionless corner frequency $v' = 4/\pi$.

The signal due to viscous drag in the x direction (Figure 25) always starts with a dilatation. The dilatation is maximum at $\theta = 90°$, $\phi = 0°$, and by symmetry at $\theta = 90°$, $\phi = 180°$. The signal due to viscous drag in the z direction (Figure 26) starts with a dilatation in the upper hemisphere ($z > 0$) and a compression in the lower hemisphere ($z < 0$). The dilatation is maximum at $\theta = 0°$, $\phi = 90°$, and the compression is maximum at $\theta = 180°$, $\phi = 90°$. The frequency content of the two types of signal is drastically different; the former (Figure 25) is characterized by a short-period oscillation, while the latter (Figure 26) is marked by a much longer period. In this example the contribution due to viscous drag is 2 orders of magnitude smaller than that due to the opening of the crack wall,

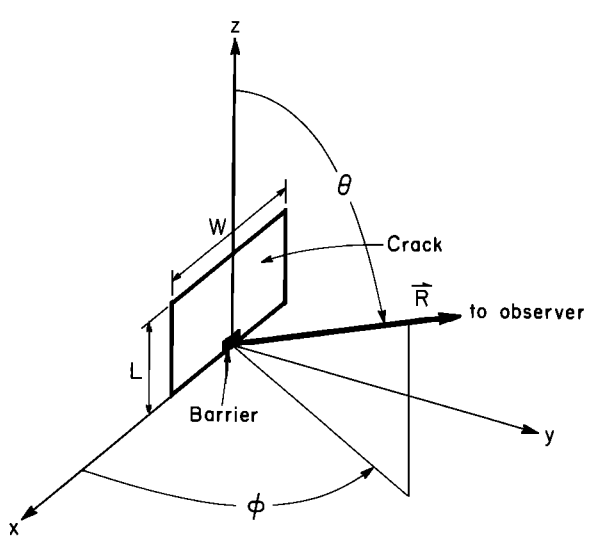

Fig. 19. Crack geometry and coordinate system used in the calculation of the far-field radiation.

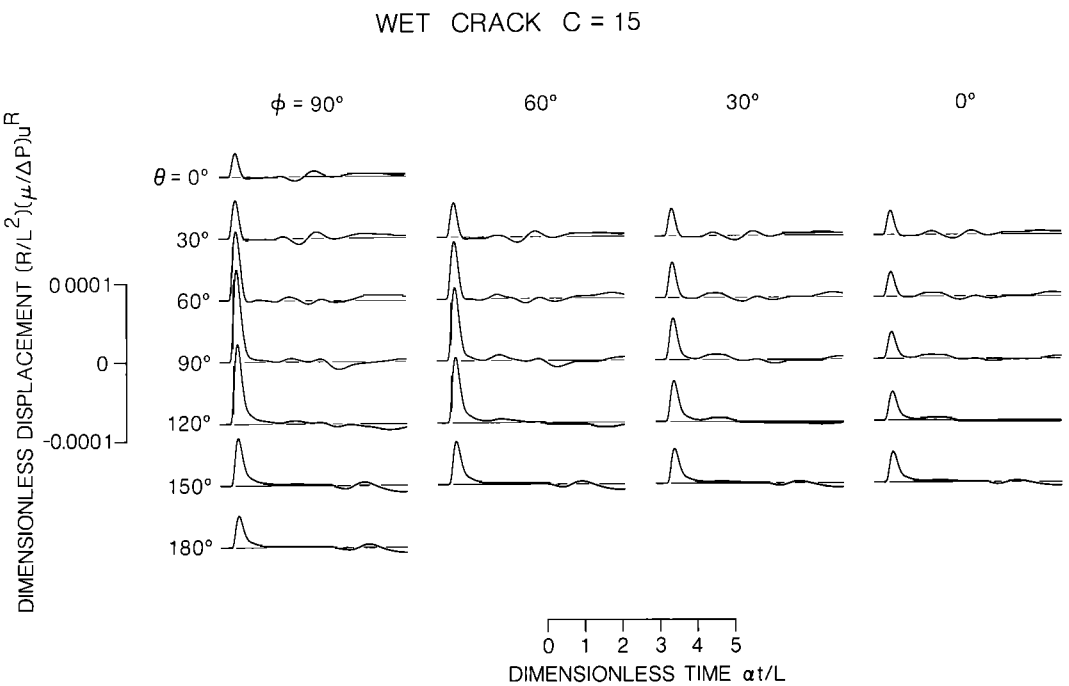

Fig. 20. Far-field compressional waveform for a fluid-filled crack with the ratio $W/L = 1$ and stiffness $C = 15$ which expands by $\Delta S = L^2/200$. The angles θ and ϕ are indicated at the left and top of the figure, respectively (refer to Figure 19). The displacement amplitudes are given in dimensionless form as a function of the crack and medium parameters (see text for details).

so the dilatational onsets seen in these figures do not affect the polarity of the first motions depicted in Figure 22. It appears that the forces resulting from the traction discontinuities across the crack faces can safely be neglected in this source model because there is no significant mass transfer within the crack. For these forces to play a role in the makeup of the far-field waveforms, the magnitude of the terms shown in equation (4) needs to be large; this may be achieved in an open system where the fluid can flow freely and the flow velocities are important.

The first motions in Figure 24 have the same polarity as those seen in Figures 20–22, and the far-field wave trains are similar to those depicted in Figure 23, although the duration of the signal is considerably reduced as a result of the increased energy loss due to viscous dissipation at the source. The long-period component of dilatation that was observed following the initial compressive pulse in Figure 23 is present at all radiation angles in Figure 24. This dilatation is most conspicuous for angles $\theta > 90°$ and appears to be characteristic of this frequency-dependent source. The overall frequency

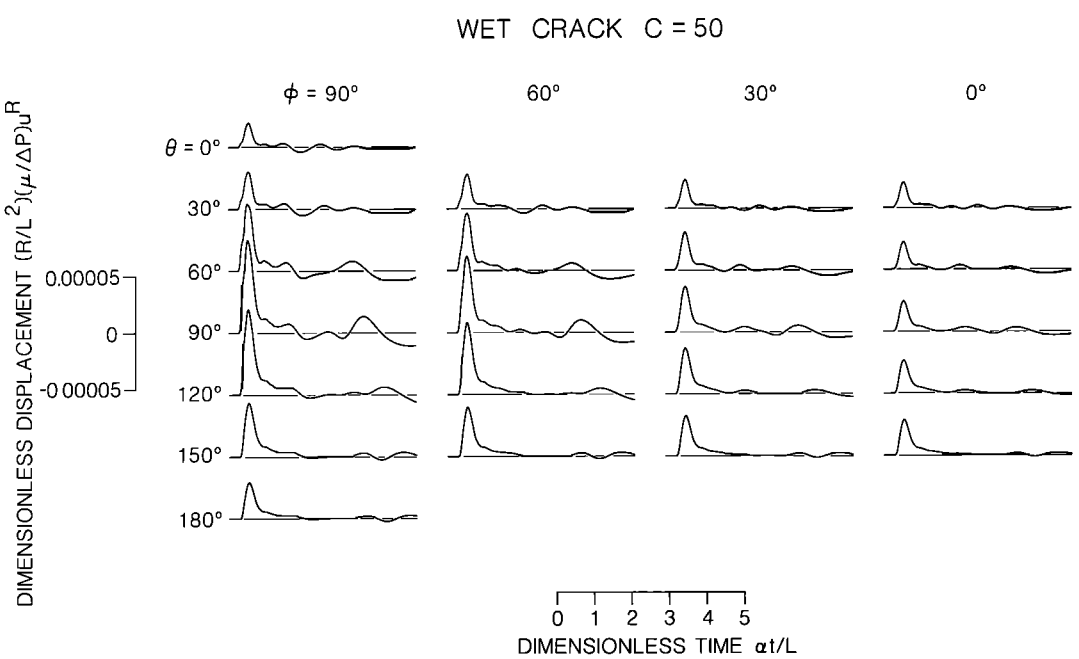

Fig. 21. Same as Figure 20 for a crack with the stiffness $C = 50$.

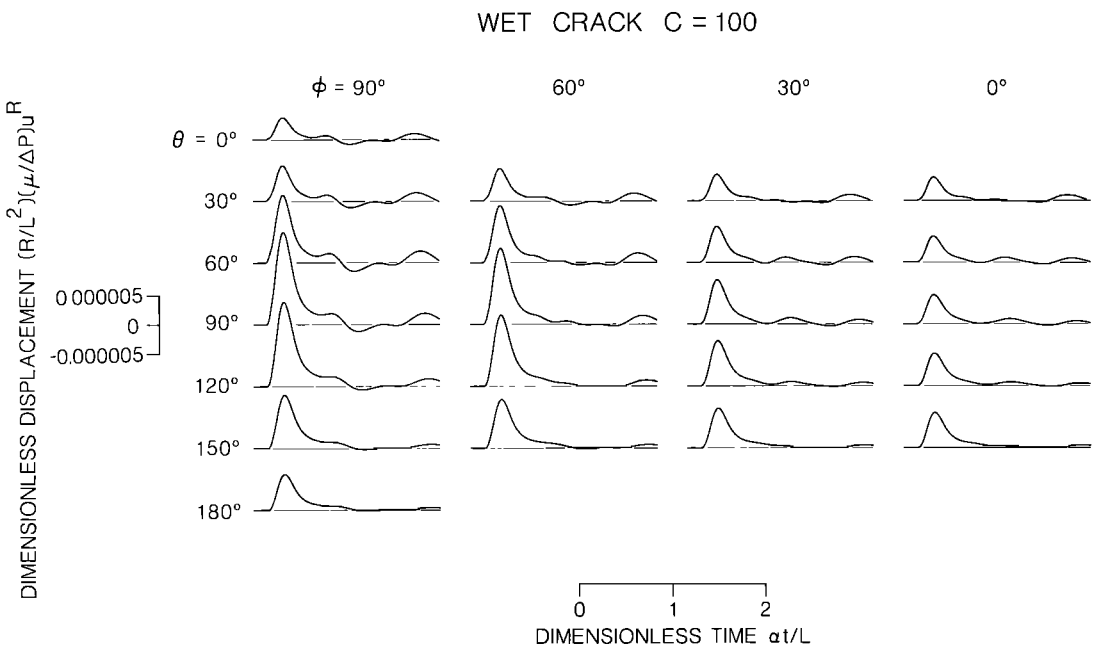

Fig. 22. Same as Figure 20 for a crack with the ratio $W/L = 0.5$ and stiffness $C = 100$ which expands by $\Delta S = L^2/800$.

content of the wave trains is fairly independent of the radiation angle, suggesting that the dominant frequency of the far-field signal may be independent of the source orientation.

Discussion

As mentioned in the introduction, the motivation for the analysis presented in this paper was provided by various recent observations of seismic events associated with man-made hydrofractures, geothermal activity, and volcanic phenomena such as magmatic injection and magmatic transport. It is of interest to discuss here what the implications of the numerical model are for some of these observations.

Three large earthquakes near Long Valley caldera in 1978 and 1980 have been found to have mechanisms with significant non-double-couple components in their seismic moment tensors [*Wallace et al.*, 1982; *Barker and Langston*, 1983; *Julian*, 1983; *Ekström*, 1983; *Ekström and Dziewonski*, 1983; *Wallace*, 1985; *Julian and Sipkin*, 1985; *Sipkin*, 1986]. These events are currently a subject of controversy, with two different explanations being proposed to account for their anomalous radiation patterns. *Wallace et al.* [1982], *Ekström* [1983], and *Ekström and Dziewonski* [1983] attribute their unusual radiation to simultaneous shear failure on fault planes of different orientations, an interpretation also favored by *Wallace* [1985], *Lide and Ryall* [1985], and *Priestley et al.* [1985] in more recent studies. *Julian* [1983], *Julian and Sipkin* [1985], and *Sipkin* [1986], on the other hand, argue that these events can be explained by a mechanism with a compensated linear vector dipole component (CLVD) [*Knopoff and Randall*, 1970] and suggest that such mechanism represents the rapid opening of a crack under high internal fluid pressure. The results of the present study are particularly relevant to this latter mechanism.

The equivalent force system for an expanding tensile crack consists of an isotropic dilatation superposed to a couple without moment acting normally to the crack plane [*Burridge and Knopoff*, 1964], while a CLVD consists of three dipoles with moments in the ratio $2:-1:-1$ [*Julian and Sipkin*, 1985]. In the first source, the resulting far-field first motions of the P wave are compressional everywhere, but in the second the first motions are dilatational in a band centered on the plane of the crack and compressional elsewhere. To generate a CLVD, an implosive component needs to be added to the tensile source that cancels out its volumetric component. *Julian* [1983] and *Julian and Sipkin* [1985] propose that this

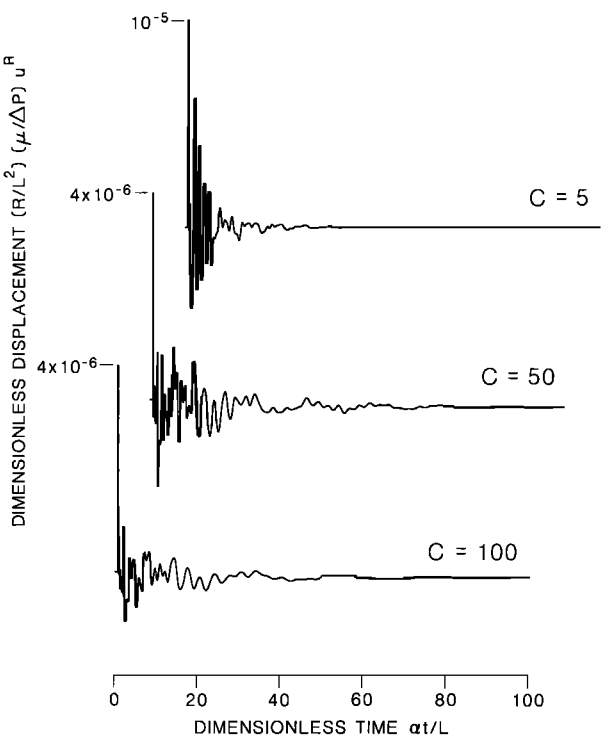

Fig. 23. Effect of crack stiffness on the far-field P wave train generated by the excitation of the crack. The crack has the ratio $W/L = 0.5$ and expands by $\Delta S = L^2/800$. The synthetics are obtained at $\theta = 30°$ and $\phi = 30°$ and are given in dimensionless form as indicated by their peak amplitudes.

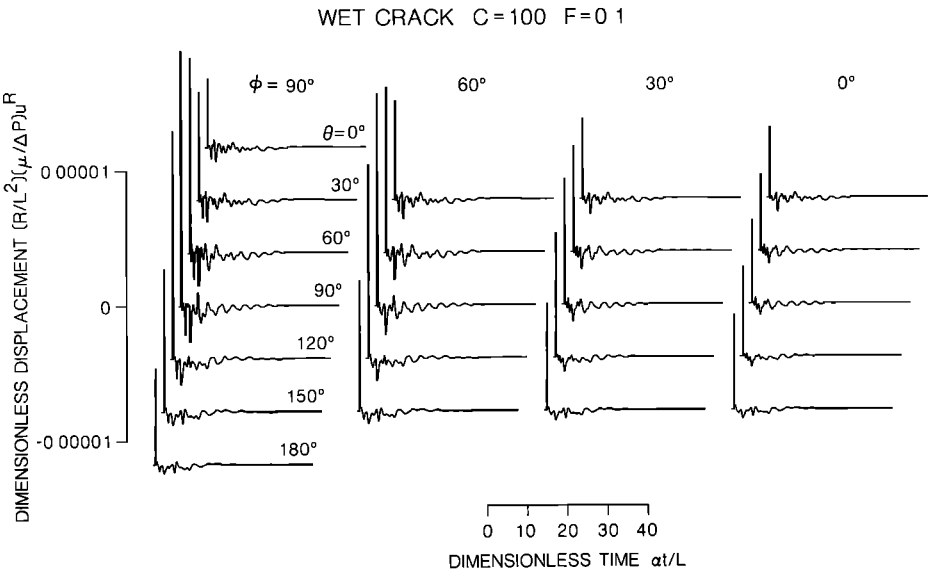

Fig. 24. Far-field P wave train generated by the excitation of a crack containing a viscous fluid characterized by $F = 0.1$. The angles θ and ϕ are the same as in Figures 20–22.

implosive component might be caused by fluid flowing into the void created by the expansion of the crack.

As shown by *Chouet and Julian* [1985] and in the present model, the drop of fluid pressure triggered by the crack expansion does indeed produce a partial collapse of the crack wall (see, for example, Figures 7, 8, 10, 11, and 12). However, since the pressure disturbance propagates in the fluid with the velocity of the crack wave, this effect is delayed with respect to the compressive pulse due to the failure of the crack. The wall collapse does produce a weak long-period component of dilatation in the far-field signal, but this dilatation does not affect the polarity of the first motion (see Figures 23 and 24). Thus the onset of the far-field P wave is always compressional independent of azimuth, and neither *Chouet and Julian* [1985] nor the model discussed here support the CLVD mechanism.

Chouet and Julian [1985] discussed various factors that may help resolve this discrepancy, including inaccuracies in the computed waveforms due to the finite thickness of the crack, anelastic attenuation along the wave path, interference between direct waves and waves reflected from the earth's surface, and the response of the recording instruments. Among these, the effect of crack thickness deserves more attention to achieve a better understanding of the radiation characteristics of a fluid-driven crack. Recent studies [*Aki et al.*, 1977; *Chouet*, 1981; *Chouet and Julian*, 1985], including this paper, use a form of the representation theorem [*Aki and Richards*, 1980] as given by equation (15), in which the variable $\Delta\dot{v}$ is considered as a discontinuity across the crack plane assuming that the crack has zero initial thickness. Of course, this is not true since the crack has a finite preexisting thickness and is filled with fluid. The above authors have assumed that (15) is valid for wavelengths $\lambda \gg d$, but the limitations of such as-

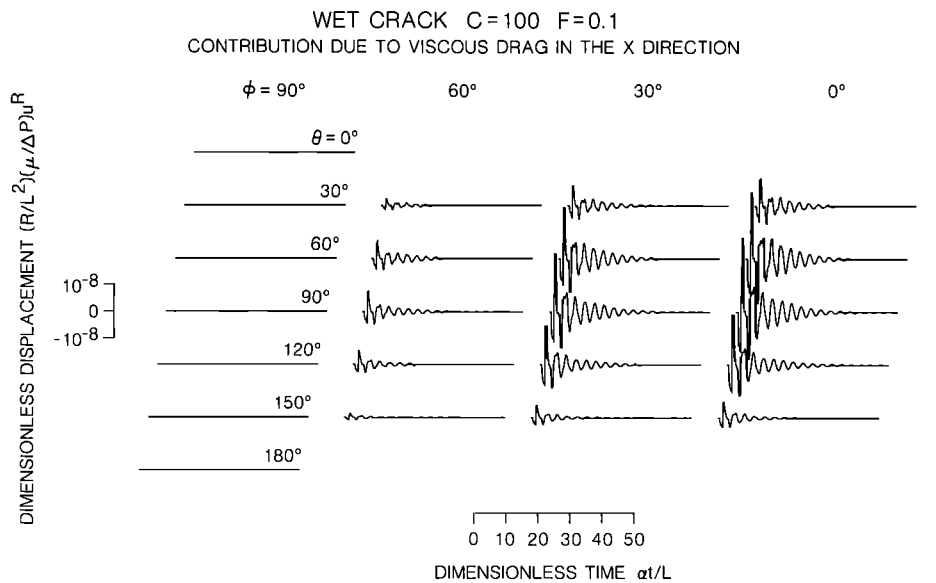

Fig. 25. Contribution to the far-field P wave train due to the viscous drag applied on the crack wall by fluid flowing in the x direction within the crack (see Figure 19 for orientation). The crack parameters and orientation of the synthetics are the same as in Figure 24.

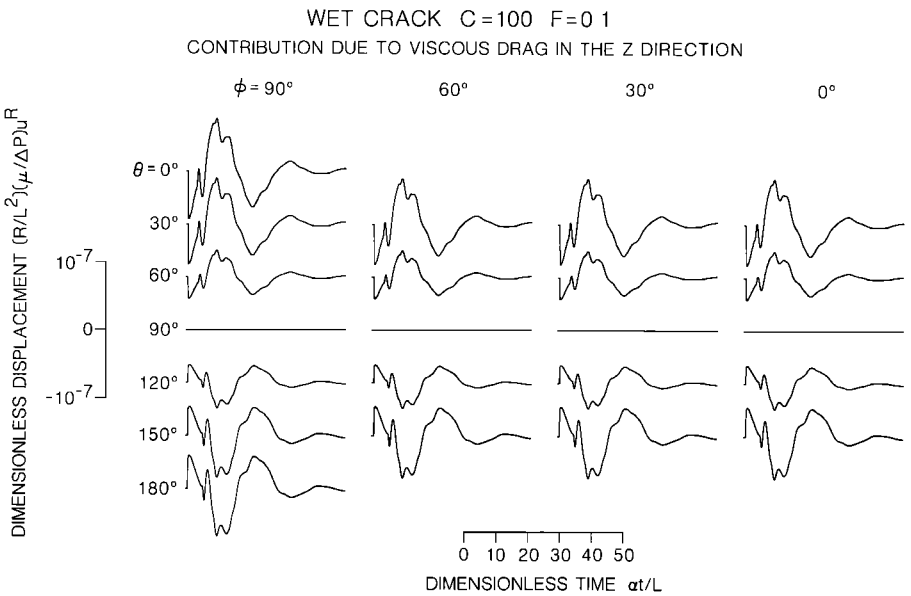

Fig. 26. Contribution to the far-field P wave train due to the viscous drag applied on the crack wall by fluid flowing in the z direction within the crack (see Figure 19 for orientation). The crack parameters and orientation of the synthetics are the same as in Figure 24.

sumption have not been tested fully yet. In particular, the effect of crack geometry and the importance of the in-plane components of wall motion along the crack perimeter in shaping the far-field signal need to be assessed quantitatively. The representation of the crack as a volume source [see *Aki and Richards*, 1980, section 3.4] may be more meaningful when dealing with thick dikes and other volcanic conduits with large cross sections.

Another important aspect of the model which is of direct relevance to the observations in the controversial Long Valley earthquakes is the spectral signature of the source. The excitation of the crack triggered by the failure of the barrier produces a source spectrum characterized by many resonant peaks (see Figures 14–16), the amplitudes of which depend on the viscosity of the fluid. The peaked nature of the source spectrum should make this type of source easily discernible from the spectral signature due to a shear-type source. As noted by *Priestley et al.* [1985], however, there is no evidence in the spectra of the Long Valley events to distinguish them from tectonic earthquakes.

Four important results derived from the present model concern (1) the dispersion characteristics of the crack wave, (2) the dependence of the crack wave velocity on the crack stiffness, (3) the resonance characteristics of the fluid-driven crack, and (4) the partition of energy between high- and low-frequency motions at the source. The first three bear directly on the estimation of the source size from the measurement of the dominant period observed in the seismic signal radiated by the fluid-driven crack. The dominant period is a critical parameter in the study of volcanic tremor [*Chouet et al.*, 1986], as well as long-period events observed in volcanoes [*Fehler and Chouet*, 1982] or in hydrofracture experiments [*Bame and Fehler*, 1986]. Because of the very low velocity which is possible in a crack with high stiffness ($C \geq 100$), the model can be used to explain observations of very long period tremor. *Kubotera* [1974] described type 2 tremor at Aso volcano with periods ranging between 3.5 and 7.0 s, which he interpreted in terms of the free oscillation of a spherical magma chamber beneath the active crater. The chamber radius compatible with the data was found to be about 2–4 km. It is clear from Table 2 that a crack with the stiffness $C \simeq 100$ a few hundred meters in length can also produce the periods required within the parameters of the above model.

The fourth result listed above is responsible for the characteristic far-field signature of the crack marked by a high-frequency onset superimposed onto a lower-frequency waveform of much longer duration. This signature is in harmony with that displayed by long-period signals observed in the field [*Fehler and Chouet*, 1982; *Bame and Fehler*, 1986].

Knowledge of the impulse response of a fluid-filled crack is a necessary prerequisite for an understanding of volcanic tremor. In relation to the magma transport problem the model discussed in this paper can be viewed as representative of the characteristic scale length of a dike between changes in its cross section or direction. All that is required to isolate dynamically a given segment of the fluid system is a boundary condition that provides a pressure wave reflection (see *Ferrick et al.* [1982] for details). With regards to the tremor process, the triggering mechanism considered in this study is quite arbitrary. There are many possible pressure disturbances that can adequately trigger the excitation of a fluid-filled crack, and boundary and initial conditions should be designed that are useful for the particular tremor-generation model being studied (i.e., a change of pressure in a reservoir, fluctuations of flow velocities due to variations in magma supply, and so on). The identification of the triggering mechanism responsible for sustained tremor remains an exciting challenge for the seismologist interested in the process of magmatic injection.

CONCLUSIONS

A finite difference method has been presented to model the dynamics of an expanding fluid-filled crack in three dimensions. The method treats the elastodynamic equations in terms of velocity and stress and the fluid dynamics in terms of flow velocity and pressure. A similar approach was used by *Chouet and Julian* [1985] in the propagation of a fluid-filled crack in two dimensions. This velocity-stress finite difference scheme has been used extensively to solve various crack and wave

propagation problems in homogeneous and heterogeneous media [*Madariaga*, 1976; *Aki et al.*, 1977; *Fehler and Aki*, 1978; *Virieux and Madariaga*, 1982; *Virieux*, 1984, 1986; *Chouet and Julian*, 1985]. The main limitations of the method are due to numerical dispersion, which limits the solution to wavelengths $\lambda > 5L/n$, and the approximate way in which the liquid and solid are connected along the crack interface (interpenetration of the fluid and solid in a layer half a grid thick over the crack surface and interpolation required to solve the fluid dynamics).

Several time scales characterize the motion of the expanding fluid-filled crack. These are the duration of motion due to the brittle failure of the barrier and the periods of acoustic resonance in the lateral and longitudinal dimensions of the crack. The failure event has a duration proportional to the area ΔS of the barrier and is the trigger that excites the resonance of the crack. The resonant periods are proportional to the crack stiffness and to the width and length of the crack; these periods are generally much longer than that of brittle failure. The crack wave is dispersive, showing a decreasing phase velocity with increasing wavelength; its wave speed also shows a strong dependence on the crack stiffness, becoming lower as the stiffness increases. The frequency content of the crack response decreases when the crack stiffness or the size of the rupture area increase. The source spectrum is marked by the presence of many narrow-band peaks that characterize the various modes of resonance of the source.

The initial motion of the crack at the rupture location is an opening. Fluid flow into the newly created cavity, however, generates a pressure drop in the fluid which triggers a partial collapse of the crack wall in the immediate vicinity of the failure.

The radiated far-field compressional wave always starts with a strong but brief compression originating at the rupture. The width of this initial pulse increases in direct proportion to the crack stiffness and incremental area of crack extension. The compressional first motion is followed by a weaker component of dilatation of much longer period, which is most conspicuous for stiffer cracks when observed in the direction of the rupture. The partition of energy between high- and low-frequency motions at the source produces a far-field signature which is enriched in high frequencies near the onset of the signal and enriched in low frequencies in the signal coda. The signal duration is strongly dependent on the viscosity of the fluid within the crack.

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